CFA® Program

Level II

FORMULA SHEET (2024) Version 1.0

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FINANCE | RISK | SUSTAINABILITY

FOR REFERENCE ONLY

(Note: Formula Sheet is not provided in the CFA exam)

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CFA Level 2 – Formula Sheet (2024)

Setting Up the Texas BA II Plus Financial Calculator

Video: https://youtu.be/0MS8d8QOFmc

QUANTITATIVE METHODS

Learning Module 1 | Basics of Multiple Regression and Underlying Assumptions

$$Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + \cdots + +b_kX_{ki} + \varepsilon_i \qquad i = 1, 2, 3, \ldots, n$$

where:
Y = dependent variable
X = independent variable
b_0 = intercept
$b_1, b_2, \ldots, b_k$ = slope coefficients
ε = error term
n = number of observations
k = number of independent variables
$b_0, b_1, b_2, \ldots, b_k$ = regression coefficients

$$Variation\ of\ Y = \sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

Learning Module 2 | Evaluating Regression Model Fit and Interpreting Model Results

Coefficient of determination, R^2

Explained Var^n / Total Variation $=$

$$R^2 = \frac{Sum\ of\ Squares\ Regression}{Sum\ of\ Squares\ Total} = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = (Multiple\ R)^2$$

$SST = SSR + SSE$

$$Sum\ of\ Squares\ Total, \quad SST = \sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

$$Sum\ of\ Squares\ Regression, \quad SSR = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$$

$$Sum\ of\ Squares\ Error, \quad SSE = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

$\because$ error $= y - \hat{y}$ & SEE $= \sqrt{\frac{SSE}{n-(k+1)}}$ → std error of the estimate

$$Adjusted\ R^2, \quad \bar{R}^2 = 1 - \left[\frac{SSE/(n-k-1)}{SST/(n-1)}\right] = 1 - \left(\frac{n-1}{n-k-1}\right)(1 - R^2)$$

→ can be derived

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AIC, BIC are useful to rank-order models. But individually, they don't have much s.f.

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Akaike's information criterion (AIC) — prefer model with lowest AIC; esp. for pred' models.

$$AIC = n \ln\left(\frac{Sum\ of\ squares\ error}{n}\right) + 2(k+1)$$

$$= n \ln\left[\frac{SSE}{n}\right] + 2(k+1)$$

where:

n = Sample size

k = Number of independent variables in the model

Schwarz's Bayesian information criterion (BIC of SBC) lowest BIC ⇒ parsimonious, best fit

$$BIC = n \ln\left(\frac{Sum\ of\ squares\ error}{n}\right) + \ln(n)\,(k+1)$$

F-distributed test statistic for jointly omitted variables

$$F = \frac{(Sum\ of\ squares\ error\ restricted\ model - Sum\ of\ squares\ unrestricted)/q}{Sum\ of\ squares\ unrestricted\ model/(n-k-1)} = \frac{(SSE^R - SSE^{UR})/q}{SSE^{UR}/(n-k-1)}$$

where:

q = Number of restrictions (i.e., number of variables omitted in the restricted model compared to the unrestricted model)

q such coeff = 0 ← $H_0: b_m = b_{m+1} = \cdots = b_{m+q-1} = 0$

H_a: At least one of the q slopes ≠ 0

F-test for joint test of slope coefficients

ANOVA	df	SS	MS	F
Regression	k	SSR	SSR/k	$\frac{SSR/k}{SSE/(n-k-1)}$
Residual	$n-k-1$	SSE	$SSE/(n-k-1)$	
Total	$n-1$	SST		

$$F\ statistic = \frac{Mean\ Square\ Regression}{Mean\ Square\ Error} = \frac{SSR/k}{SSE/(n-k-1)}$$

$H_0: b_1 = b_2 = \cdots = b_k = 0$ // same as above

H_a: At least one $b_j \neq 0$

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t-test statistic for slope coefficient

may extract s/e from existing regn output s.t. s/e = coeff / t-val & then recalc t-stat for diffnt null abt the val of the coeff.

$$t = \frac{\hat{b}_j - B_j}{s_{\hat{b}_j}}$$

where:

$\hat{b}_j$ = Regression estimate of b_j

B_j = Hypothesized value of coefficient j

$s_{\hat{b}_j}$ = Estimated standard error of $\hat{b}_j$

Video (Simple Linear Regression): https://youtu.be/uR_9im2JP18

Learning Module 3 | Model Misspecification

Breusch-Pagan Test

$$\text{Test Statistic, } \quad \chi^2_{BP,k} = nR^2$$

Rem

where:

this is diffn from R^2 @ Pg 2

R^2 = R-squared between squared residuals (not the DV) and independent variables → test heteroskd; Soln = robust s/e for stat. infrn ← is a problem if conditional on X

Variance Inflation Factor (VIF)

$$VIF_j = \frac{1}{1 - R_j^2}$$

where:

another version of R^2 → varies w j

R_j^2 = Variation in X_j explained by the other $k - 1$ independent variables

Learning Module 4 | Extensions of Multiple Regression

Conditional heterosk ⇒ var(ϵ) is ~~not~~ correlated with values of Indep. Vars.

Detecting Influential Points

Sum of individual leverages for all observations = $k + 1 = \sum h_i$

✓ If observation's leverage > $3\left(\frac{k+1}{n}\right)$ ⇒ Potentially influential observation Test ①

Studentized Deleted Residual, t_{i^*}

$$t_{i^*} = \frac{e_i^*}{s_{e^*}} = \frac{e_i}{\sqrt{MSE_{(i)}(1 - h_{ii})}} \sqrt{\frac{n - k - 1}{SSE(1 - h_{ii}) - e_i^2}}$$

where:

e_i^* = The residual with the ith observation deleted

s_{e^*} = The standard deviation of the residuals

k = The number of independent variables

$MSE_{(i)}$ = Mean squared error of the regression model that deletes the ith observation

h_{ii} = The leverage value for the ith observation

Multi Collin: when R^2 is High but insf t-stat on Indep Vars. ; also detected & quantified by VIF

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Type I error: Reg when Null is true || Breusch-Pagan test ... is heteroskedasticity.
Type II " : fail to reg " " " False ↳ calc robust s/e



Cook's Distance

$$D_i = \frac{e_i^2}{(k+1)MSE}\left[\frac{h_{ii}}{(1-h_{ii})^2}\right]$$

where:

e_i = Residual for observation i

k = The number of independent variables

MSE = Mean square error of the estimated regression model = SSE/(n-k-1)

h_{ii} = The leverage value for the ith observation

If $D_i > \sqrt{k/n}$, then ith observation is highly likely to be an influential data point ⟶ Test (2)

Logistic Regression (Logit)

$$\text{log(odds ratio)} = \ln\left(\frac{P}{1-P}\right) = b_0 + b_1X_1 + b_2X_2 + \cdots + b_kX_k + \varepsilon$$

$$P = \frac{1}{1+\exp[-(b_0 + b_1X_1 + b_2X_2 + \cdots + b_kX_k + \varepsilon)]}$$

$$\ln\left(\frac{P}{1-P}\right) = Log\ odds$$

$$Odds\ ratio = e^{b_i}$$

Likelihood ratio (LR) test

$$LR = -2(Log\ likelihood\ restricted\ model - Log\ likelihood\ unrestricted\ model)$$

Learning Module 5 | Time-Series Analysis

Linear Trend Models

$$Y_t = b_0 + b_1t + \varepsilon_t \qquad t = 1,2,\ldots,T$$

Reqm to check if t is out-of-sample.

t = time (independent variable)

Log-Linear Trend Models

$$Y_t = e^{b_0+b_1t} \qquad t = 1,2,\ldots,T$$

$$\ln Y_t = b_0 + b_1t$$

✓ Growth rate of $Y = e^{b_1} - 1$

Time series is cov-stat" if properties (μ,σ) do not change over time. i) $E(\mu)$ = const, finite $\forall t$ ii) σ is const, finite $\forall t$; 3) cov. of t/s w itself for a fixed # lags or leads be const & finite $\forall t$.

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Durbin-Watson (DW) test :- H_0: errors r not serially correlated. DW = 2 ⇒ no auto corr; DW < 2 ⇒ +ve Ser. Corr. DW > 2 ⇒ −ve Ser. Corr.

p-th order autoregressive model, $AR(p)$

$$x_t = b_0 + b_1 x_{t-1} + b_2 x_{t-2} + \cdots + b_p x_{t-p} + \varepsilon_t$$

Test statistic for autocorrelation of residuals

$$t = \frac{\text{Residual autocorrelation} - 0}{\text{Standard error}} = \frac{\text{Residual autocorrelation}}{1/\sqrt{T}} = \frac{\rho_{\varepsilon,k}}{1/\sqrt{T}}$$

where:

$$\rho_{\varepsilon,k} = \frac{Cov(\varepsilon_t, \varepsilon_{t-k})}{\sigma_\varepsilon^2} = k^{th} \text{ order autocorrelation of the residual}$$

Mean reverting level for AR(1) model ⟶ cov. stationary

$$x_t = \frac{b_0}{1 - b_1}$$

Root Mean Squared Error

$$RMSE = \sqrt{\frac{\text{Squared error}}{n}}$$

Dickey and Fuller Unit-Root Test

$$x_t - x_{t-1} = b_0 + g_1 x_{t-1} + \varepsilon_t$$

where:

$g_1 = b_1 - 1$ Test if $g_1 = 0$, i.e. $b_1 = 1$ → H_0: $g_1 = 0$ → t/s has unit root & non-stat^n i.e. r.w like
H_A: $g_1 < 0$ → t/s does NOT have unit root & is stat^n
∴ If p-val not s.f. ⇒ H_0 ✓ → unit rt ⟶ non-stat^n

ARCH(1):

$$\hat{\sigma}_{t+1}^2 = a_0 + a_1 \hat{\epsilon}_t^2$$

Learning Module 6 | Machine Learning

Neural Networks

$$\text{New network weight} = \text{Old weight} - \text{Learning rate} \times \left(\begin{array}{c} \text{Partial derivative of the} \\ \text{total error with respect} \\ \text{to the old weight} \end{array} \right)$$

Discrim. Analysis – yields a linear fn. ≃ ~~linear~~ regn. eqn. → can be used to score Y; then classify as 1 or 0.
Agglomerative clustering: unsupv., hierarchical clustering → starts with N clusters & progressively iterates to end w. 1 cluster
K-means clustering – is well-suited for hi-dimens data.

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Dimension = focus attention on a few imp. vars i/o selecting some probabilistic variables.

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Learning Module 7 | Big Data Projects

range ← Normalization of variable X

$$X_{i\,(normalized)} = \frac{X_i - X_{min}}{X_{max} - X_{min}}$$

Standardization of variable X

$$X_{i\,(standardized)} = \frac{X_i - \mu}{\sigma}$$

$$Precision, P = \frac{True\ Positive}{True\ Positive + False\ Positive} = \frac{TP}{TP+FP}$$

$$Recall, R = \frac{True\ Positive}{True\ Positive + False\ Negative} = \frac{TP}{TP+FN}$$

$$Accuracy = \frac{True\ Positive + True\ Negative}{True\ Positive + False\ Positive + True\ Negative + False\ Negative}$$

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

switch both

$$False\ Positive\ Rate, FPR = \frac{False\ Positive}{True\ Negative + False\ Positive}$$

$$True\ Positive\ Rate, TPR = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

$$Document\ Frequency, DF = \frac{Sentence\ Count\ with\ Word}{Total\ Number\ of\ Sentences}$$; Term Freq (TF) = $\frac{\#\,(term\ in\ doc)}{Total\ \#\ terms\ in\ doc}$

$$Inverse\ Document\ Frequency, IDF = \log\left(\frac{1}{DF}\right)$$

$$TF\text{-}IDF = TF \times IDF$$

Inconsistency error = where data conflicts w correspdg data pts or reality.

Aggregation (summarizing) & hot-coding are transformal processes. → creating new features

Tradeoff blw bias error vls variance error plotted on Fitting curve. Learning Curve = pattⁿ of convgnc blw training accu & validⁿ accur as # training samples ↑. Hi bias error ⇒ Underfitting; vls Var error measures how well the model fits new data from validⁿ & testing samples.

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Fwd disc/prem = $\left(\frac{F-S}{S}\right) \times \frac{12}{\#mth} \times 100$

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ECONOMICS

Learning Module 1 | Currency Exchange Rates: Understanding Equilibrium Value

Cross Rates

When given $\frac{A}{C}$ and $\frac{C}{B}$, then $\frac{A}{B} = \frac{A}{C} \times \frac{C}{B}$

When given $\frac{A}{C}$ and $\frac{B}{C}$, then $\frac{A}{B} = \frac{A}{C} \times \frac{1}{\left(\frac{B}{C}\right)}$

Currency pair	Bid	Bid/Ask
A/B	x	y
B/A	$1/y$	$1/x$

Video: https://youtu.be/wyDKKPkPhzw

Arbitrage Opportunities Between Dealers and Interbank

Video: https://youtu.be/Lqo9UZ3yyEA

Covered Interest Rate Parity – yld of a fore money mkt instru, hedged for exch rate risk is = yld of domes money mkt instru.

Forward Rate $$F_{f/d} = S_{f/d}\left[\frac{1 + i_f\left(\frac{Actual}{360}\right)}{1 + i_d\left(\frac{Actual}{360}\right)}\right] \longrightarrow (1)$$

derive from (1) ← $$\frac{F_{f/d} - S_{f/d}}{S_{f/d}} = \frac{(i_f - i_d)\left(\frac{Actual}{360}\right)}{1 + i_d\left(\frac{Actual}{360}\right)}$$

Video: https://youtu.be/9jOzFA9GuHU

Mark-to-Market Value of a Forward Contract for ccy pair

Original position: **Long** base currency *d* forward at forward rate $F_{0,f/d}$ (Offer side) → buying fwd @ Ask @ t = 0

$$Value\ of\ Long\ Forward = \frac{\left(F_{t,f/d} - F_{0,f/d}\right) \times Contract\ Size}{1 + i_f\left(\frac{Remaining\ days\ to\ maturity}{360}\right)}$$

$F_{t,f/d}$ = Forward rate at valuation date, t (Bid side) → unwind on opp. side

Video: https://youtu.be/wLqyZRrutAc

Both fwd & spot bid/ask depend on 1) IB mkt liquidity; 2) size of trxn 3) r-ship blw client & dealer

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If both CIP & UCIP hold, then fwd exch rate over a term will be an unbiased predictor of spot after term.

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Borrowing ccy is Numr in Carry Trade

Uncovered Interest Rate Parity very similar to CIP ; MUST not Hold for carry Trade to ret" profit

Expected future spot Rate: $E(S_{f/d}) = S_{f/d}\left[\frac{1+i_f\left(\frac{Actual}{360}\right)}{1+i_d\left(\frac{Actual}{360}\right)}\right]$

$$\%\Delta S^e_{f/d} = \frac{(i_f - i_d)\left(\frac{Actual}{360}\right)}{1+i_d\left(\frac{Actual}{360}\right)} \approx (i_f - i_d)\left(\frac{Actual}{360}\right) \quad \underline{OR} \quad \%\Delta S^e \simeq R_A - R_B$$

Video (Carry Trade): https://youtu.be/_26fG3Zvzyg

↳ benefits when hi-yld ccy apprec → 1) $(i_H - i_L)\uparrow$; 2) $(\Psi_H - \Psi_L)\downarrow$ Risk Prem; 3) $(\pi^e_H - \pi^e_L)\downarrow$

Absolute PPP

$$S_{f/d} = \frac{P_f}{P_d}$$

Relative PPP → via inflation & ≃ UCIP

$$\%\Delta S_{f/d} = \frac{(\pi_f - \pi_d)\left(\frac{Actual}{360}\right)}{1+\pi_d\left(\frac{Actual}{360}\right)} \approx (\pi_f - \pi_d)\left(\frac{Actual}{360}\right)$$

Ex ante PPP - expected inflation

$$\%\Delta S^e_{f/d} \approx (\pi^e_f - \pi^e_d)\left(\frac{Actual}{360}\right)$$

International Fisher Effect OR compare real rate in A vs B & invest higher

$$i_f - i_d = \pi^e_f - \pi^e_d$$

where:

π^e = Expected inflation rate

π = Actual inflation rate

i_f, i_d = Nominal Int rate in foreign & domestic mkts.

Mundell-Fleming Model

Bonus Video: https://youtu.be/xNo3GpWYgKA

Warning signs for ccy crisis : 1) Large infl of forein capital (vs GDP) 2) Fxd or partially fxd exchg rates 3) Broad money growth & ratio of M2/bank reserv ↑↑

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$R_f \propto$ growth in GDP (real)] ∴ higher gld needed to encourage investors to save for future consumption.
$R_f \propto \sigma$ (GDPgrowth, real)

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Learning Module 2 | Economic Growth and the Investment Decision

$$P = GDP \times \left(\frac{E}{GDP}\right) \times \left(\frac{P}{E}\right)$$

Return on aggregate equity market

$$\%\Delta P = \%\Delta GDP + \%\Delta\left(\frac{E}{GDP}\right) + \%\Delta\left(\frac{P}{E}\right)$$

where:
P = aggregate value of equities
E = aggregate earnings

Cobb-Douglas Production Function

$$Y = TK^{\alpha}L^{1-\alpha} \qquad \text{where } \alpha < 1$$

where:
Y = Output
α = Share of output allocated to capital (K)
$1-\alpha$ = share of output allocated to labor (L) Developing country: $(1-\alpha)$ is large; ⇒ low $\left(\frac{K}{L}\right)$ vis developed.
T = total factor productivity (TFP), represents technological progress of the economy

Output per worker $= \frac{Y}{L} = T\left(\frac{K}{L}\right)^{\alpha}$ ⇒ "Capital Deepening"; if K/L is already v. hi (eg. devlpd), more capital deepening won't increase Y/L. So devlpd countries rely on tech progress via TFP=T

Marginal product of capital, MPK $= \frac{\partial Y}{\partial K}$

$$MPK = \alpha\left(\frac{Y}{K}\right) = r = \text{reqd rate of ret' for capital}$$

→ Amount of output that is allocated to providers of capital, a <share>

$$\alpha = \frac{rK}{Y}, \text{ where } r = MPK$$

Ranking emrg countries (If country has lowest α ⇒ highest $(1-\alpha)$)

Growth Accounting equation:

$$\frac{\Delta Y}{Y} = \frac{\Delta T}{T} + \alpha\frac{\Delta K}{K} + (1-\alpha)\frac{\Delta L}{L}$$

these 2 maybe used togthr to rank-order countries

Growth rate in potential GDP = Long-term growth rate of **labor force** + Long-term growth rate in **labor productivity**

$$Labor\ force\ participation = \frac{Labor\ force}{Working\ age\ population}$$

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Dutch disease: country w hi resources → strong dem → ccy ↑↑ // export ⇒ other segmt like mfg is uncompetitive

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4 a) **Sustainable growth rate of output per capita**

$$g^* = \frac{\theta}{1-\alpha}$$

b) **Sustainable growth rate of output (Steady state growth rate)** ; $g(K/L) = g(Y/L)$ <growth rates>

$$G^* = \frac{\theta}{1-\alpha} + n$$

c) **Equilibrium** output-to-capital ratio (in **steady state**):

$$\frac{Y}{K} = \frac{1}{s}\left[\frac{\theta}{1-\alpha} + n + \delta\right]$$

→ MPK = r; capital is just being paid its reqd rate of ret.
→ capital deepening does not affect GDP growth

→ $g\left(\frac{\Delta K}{L}\right)$ & $g\left(\frac{\Delta Y}{L}\right) = \frac{\theta}{1-\alpha}$ = const.

where:

θ = growth rate of TFP ✓ = $\Delta A/A$

α = elasticity of output with respect to capital ✓

s = fraction of income (Y) that is saved

δ = rate of depreciation of physical capital stock

n = labor supply growth = $\%\Delta L$ = $\frac{\Delta L}{L}$

7. **Endogeneous Growth Model**

Production function:

$$y_e = ck_e$$

Growth rate of output per capita: → long term potential real GDP grw rate

$$\frac{\Delta y_e}{y_e} = \frac{\Delta k_e}{k_e} = sc - \delta - n$$

where:

y_e = output per worker

k_e = capital per worker (endogns)

c = marginal product of capital in the aggregate economy (constant)

Fiscal budget deficit ↑↑ when actual GDP grw < potential GDP grw & vice-versa

State backed govt agencies or indep regulatr v/s

Indus SRegu body = pvt org that represent & regulate

↳ SRO → : get recog.n. & authority + enforcemt pwr by govt body/agency. ; funded indep.

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NI to parent S/H same b/w partial vis full GW; but RoA & RoE will be diffnt.

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FINANCIAL STATEMENT ANALYSIS

Learning Module 1 | Intercorporate Investments

Investments in Associates (Equity Method)

$$\text{Ending investment in associates} = \text{Beginning investment in associates} + \left[\text{Share of net income} - \text{Share of dividend received} - \text{Amortization of excess purchase price}\right]$$

Impact on **Investor's** Income Statement

$$= \text{Share of net income} - \text{Amortization of excess purchase price} - \text{Share of Unrealized profit from downstream or upstream sale}$$

Business Combinations (Acquisition Method)

Excess purchase price

= Acquisition price – %Ownership × **Book** value of net identifiable assets

Partial Goodwill – focussed on net identifiable assets & acqsn. Px → = Ⓐ – Debt

= Acquisition price – (%Ownership × **Fair** value of identifiable net assets)

= Acquisition price – (%Ownership × **Book value** of identifiable net assets)

– (%Ownership × Excess purchase price attributable to identifiable net assets)

$$\text{Non controlling interest} = \%NCI \times \text{Fair value of identifiable net assets}$$

Full Goodwill → reqd by US Gaap.

= Fair value of entity – **Fair** value of net identifiable assets

$$\text{Non controlling interest} = \%NCI \times \text{Fair value of entity}$$

When Control is observed & be present

Video: https://youtu.be/RgxmPbx4-0o

IFRS → Full or Partial GW allowed.

$$\text{Impairment loss} = \text{Carrying value of cash generating unit} - \text{Recoverable amount of cash generating unit}$$

US GAAP → only allow full goodwill when there is non-ctrlg intr in affiliate

$$\text{Implied goodwill} = \text{Fair value of reporting unit} - \text{Fair value of reporting unit's identifiable net assets}$$

$$\text{Impairment loss} = \text{Carrying value of goodwill} - \text{Implied goodwill}$$

When investor acquires ~~part~~ of Associate, any excess consid" above FV of net assets is → GW → Not amortized.
~~IFRS~~ i) excess paid = Purch Px – % of BV; out of excess, GW = Purch Px – % of FV of net Ⓐ. Than non-GW excess gets amortized from Investee's profit recognized on I/S.

F/S are decision-useful if info is relevant & faithful repⁿ of eco. reality.

IFRS: hyperinfla"; translate cash bal in a single step @ current y/e rate ∴ cash doesn't require restmt b4 Trans lation; whereas other monetary Ⓐ & Ⓛ are first restated for inflation & then convert by exch. rate → (14)

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Compen. Exp. = Options granted × Option Px on grant date ÷ 5-year service period

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Learning Module 2 | Employee Compensation - Post-Employment and Share-Based

Share-Based Compensation Accounting

→ sh-based expense impacts SH equity, not Ⓛ. <via ~~SH~~ equity reserve acc.>

$$\text{Compensation expense} = \frac{\text{Fair value of award on grant date}}{\text{Vesting period}}$$

Treasury Stock Method

$$\text{Diluted shares outstanding} = \text{Basic shares outstanding} + \text{Shares issued from conversion or exercise of share based awards} - \frac{\left(\text{Proceeds from conversion or exercise of share based awards}\right)}{\text{Average share price for the reporting period}}$$

$$\text{Assumed proceeds} = \text{Cash proceeds from exercise} + \text{Average unrecognized share-based compensation expense}$$

Forecasting Shares Outstanding With Share-Based Awards

$$\text{Basic shares outstanding, end of period} = \text{Basic shares outstanding, beginning of period} + \text{RSUs vested} + \text{Share options exercised} + \text{Shares issued from secondaries, acquisitions, etc.} - \text{Share repurchases}$$

$\text{Plan Ⓐ}^{e} = \text{Plan Ⓐ}^{b} \times (1 + \text{actual ret}^{n}) + \text{Contri} - \text{Benefit Paid}$

→ only impact fair pension plan of CF statement
↳ shown as CFO [Ops]

Financial Reporting for Defined Benefit Pension Plans

$$\text{Funded status} = \{\text{Fair value of plan Ⓐ assets} - \text{Pension obligation (PVDBO)}\}^{y/end}$$

$$\text{Ending fair value of plan assets} = \text{Beginning fair value of plan assets} + \text{Contributions} + \text{Actual return on plan assets} - \text{Benefits paid}$$

↳ depends on plan formula

$$\text{Ending pension obligation} = \text{Beginning pension obligation} + \text{Service costs} + \text{Interest cost} + \text{Actuarial loss/(gain)} - \text{Benefits paid}$$

$$= \text{PVDBO}^{b} \times (1 + \text{Disc Rate}) + \text{Benefit Earned} - \text{Benefit Paid}$$

$$\text{Interest cost} = \text{Beginning pension obligation} \times \text{Yield on investment grade corporate bond}$$

Total Periodic Pens Cost = Curr service cost + Int cost + Past service cost − Actuarial Gain + Actuarial Loss − Actual Ret^{n} on Ⓐ

IFRS Only

$$\text{Net interest cost/(income)} = \left(\text{Beginning pension obligation} - \text{Beginning fair value of plan assets}\right) \times \text{Yield on investment grade corporate bond}$$

↳ = discount rate

US GAAP: Actual ret^{n} − Exp. ret^{n} ⇒ ∝ I (for pension assets) & subsequently amortized in P&L

DBP plans regd to pre-fund plan ∵ govt partially insures; but not for OPB ; OP other benefits are not legally regd to prefund

Estimating future healthcare costs (aft retirmt) is more difficult than estimating future retirement benefits.

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Temporal Meth: If subsi Mone Ⓐ > Mone Ⓛ & foreign ccy ↑ ⇒ CTA gain
" " " Ⓐ < " Ⓛ & " " ↓ ⇒ " loss/gain
→ CTA/L added to PnL. " " " ↑ ⇒ CTA Loss

→ Temporal when f(x) ccy of subsi is same as Parent's foreign ccy & parent ccy

vis current Meth ↓
use Tot Ⓐ vis Tot Ⓛ

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Functional ccy = f [ccy for cash; ccy for sale p/t; country affecting regulⁿ & competⁿ for sales]

Learning Module 3 | Multinational Operations

Net assets = Total assets – Total liabilities

Net monetary assets = Monetary assets – Monetary liabilities

→ when subsi has foreign ccy as functional ≠ Parent's f(x) ccy.

Current Rate Method – record CTA as OCI → viz part of SH equity. ; use y/end rate for B/S items

→ If Net Ⓐ > 0 & foreign ccy ↓ → higher val of Ⓐ translated as a gain

Currency translation adjustment
= Total assets of foreign subsidiary in parent currency terms
– Total liabilities of foreign subsidiary in parent currency terms
– Shareholder capital of foreign subsidiary in parent currency terms
– Other equity items of foreign subsidiary in parent currency terms

$\left[TA - TL - (SH + \text{other eq})\right]^{\text{par ccy}}$

Fcy exposure arises due to 1) ccy diff & 2) timing diff ; if imp/exp in fcy is settld on purch dt → No expos

Hyperinflationary Environment

IFRS

$$\text{Restatement factor for monetary assets \& liabilities} = \frac{\text{End of period price index}}{\text{End of period price index}}$$

$$\text{Restatement factor for monetary assets \& liabilities} = \left[\frac{\text{End of period price index}}{\text{Beginning of period price index}}\right]^{B/S}$$

$$\text{Restatement factor for income statement items} = \left[\frac{\text{End of period price index}}{\text{Average price index for the period}}\right]_{PnL}$$

Learning Module 4 | Analysis of Financial Institutions

$$\text{Total Tier 1 Capital} = \text{Common Equity Tier 1 Capital} + \text{Additional Tier 1 Capital}$$

$$\text{Total Regulatory Capital} = \text{Total Tier 1 Capital} + \text{Total Tier 2 Capital}$$

$$\text{Common Equity Tier 1 Ratio} = \frac{\text{Common Equity Tier 1 Capital}}{\text{Risk Weighted Assets}} \geq \boxed{4.5\%}$$

$$\text{Tier 1 Ratio} = \frac{\text{Total Tier 1 Capital}}{\text{Risk Weighted Assets}} \geq \boxed{6.0\%}$$

$$\text{Total Capital Ratio} = \frac{\text{Total Regulatory Capital}}{\text{Risk Weighted Assets}} \geq \boxed{8.0\%}$$

Both IFRS & US Gaap ⇒ require disel of 1) ΔG/L due to each rate on NI, 2) cum trans adjm in Shd of Equity, 3) Recm of ΔCTA since last period ; US Gaap also requires disclosing any CTA transfrd from Eq to NI.

Disappearing Plant Problem: carrying assets in hyperinflat envmt @ histo cost & translating to parent ccy @ CR (current)
∴ US Gaap: requires translatⁿ @ Histo rate, if hyp infla. IFRS: FA carried @ Inflatⁿ adj cost & translate to CR.

Temporal Method: only monet Ⓐ, Ⓛ translated @ curr. exchg rate; non Mone Ⓐ, Ⓛ trans @ histo rate. Monetary items = cash, receivbl, payable. → fxd ccy units. Non-Monet = inv, F.A., intangible; Non-M Ⓛ = deferd rev.

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Bank earnings more sustainable when reliance on trading Rev (vis NIM rev & service income)
Provsn for loan loss = Expns a/c ; Inc => Greater Allwn for loan losses vis contra asset a/c on B/S

$$\text{Liquidity Coverage Ratio, LCR} = \frac{\text{High Quality Liquid Assets}}{\text{Expected cash outflows}}$$

$$\text{Number of days that bank can withstand a stress level volume of cash outflows} = LCR \times 30$$

Number of days that bank can withstand a stress level volume of cash outflows for $(LCR \times 30)$ days.

$$\text{Net Stable Funding Ratio, NSFR} = \frac{\text{Available Stable Funding}}{\text{Required Stable Funding}}$$

Property and Casualty Companies Reins cos have highest systemic risk (vis P&C, L&H) ∵ interntl linkages to fin instns → ∴ imp to global fin system.

Note the diffn in Denominators

$$\text{Loss and loss adjustment expense ratio} = \frac{\text{Loss expense} + \text{Loss adjustment expense}}{\text{Net premiums earned}}$$ → degree of success in UW estimating the risks insured. Low ✓

$$\text{Underwriting expense ratio} = \frac{\text{Underwriting expense}}{\text{Net premiums written}}$$ → % of money spent in obtaining new prems.; Low ✓

$$\text{Combined ratio} = \text{Loss and loss adjustment expense ratio} + \text{Underwriting expense ratio}$$ → overall % of money spent to get new prem
↳ the lower => more success.

$$\text{Dividends to policyholders (shareholders) ratio} = \frac{\text{Dividends to policyholders (shareholders)}}{\text{Net premiums earned}}$$

$$\text{Combined ratio after dividends} = \text{Combined ratio} + \text{Dividends to policyholders (shareholders) ratio}$$

Learning Module 5 | Evaluating Quality of Financial Reports

Beneish Model

$$M\text{-score} = -4.84 + 0.920\,(DSR) + 0.528\,(GMI) + 0.404\,(AQI) + 0.892\,(SGI) + 0.115\,(DEPI) - 0.172\,(SGAI) + 4.670\,(Accruals) - 0.327\,(LEVI)$$

where:

$$\text{DSR (day sales receivable index)} = \frac{Receivables_t / Sales_t}{Receivables_{t-1} / Sales_{t-1}}$$

$$\text{GMI (gross margin index)} = \frac{GM_{t-1}}{GM_t}$$

$$\text{AQI (asset quality index)} = \frac{[1 - (PPE_t + CA_t)/TA_t]}{[1 - (PPE_{t-1} + CA_{t-1})/TA_{t-1}]}$$

$$\text{SGI (sales growth index)} = \frac{Sales_t}{Sales_{t-1}}$$

Equity Method: if ≥ 20% <influence> ownership in investee → share of earning to parent = (EPS - DPS) * #sh.
FV thr PL (FVPL): lower ownership ∴ Tot Incom = (Div + cap. gain due to sh Px ↑) * # sh.

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May be IFRS compliant, but if Int Incm & Div Incm as Op Activ (vs Inv) → & int exp as Fin Activ → the selective, inconsistent choices indicate potential warning abt c/f activity.

Noesis

$$\text{DEPI (depreciation index)} = \frac{Depreciation_{t-1}}{Depreciation_t}$$

$$\text{SGAI (sales, general, and administrative expenses index)} = \frac{SGA_t/Sales_t}{SGA_{t-1}/Sales_{t-1}}$$

$$\text{Accruals} = \frac{Income\ before\ extraordinary\ items - Cash\ from\ operations}{Total\ assets}$$

$$\text{LEVI (leverage index)} = \frac{Leverage_t}{Leverage_{t-1}}$$

c/f	IFRS	US GAAP
Intr +	Op / Inv	Op
Intr −	" / Fin	Op
Div +	Op / Inv	"
Div −	Op / Fin	Fin

Earnings Persistence

$$Earnings_{t+1} = \alpha + \beta(Earnings_t) + \varepsilon$$

$$Earnings_{t+1} = \alpha + \beta_1(Cash\ flow_t) + \beta_2(Accruals_t) + \varepsilon$$

Cash-flow-based accruals = NI – (CFO + CFI)

If g(NI) > g(op c/f) ⇒ earnings grew due to accruals & ∴ there is rapid reversion to mean earnings expected.

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Learning Module 6 | Integration of Financial Statement Analysis Techniques

$$\begin{aligned} \text{Net Operating Assets (NOA)} &= \text{Operating Assets} - \text{Operating Liabilities} \\ &= \left(\text{Total Assets} - \text{Cash and Short-term Investments}\right) - \left(\text{Total Liabilities} - \text{Total Debt}\right) \end{aligned}$$

$$\text{Balance-sheet-based accruals ratio} = \frac{NOA_t - NOA_{t-1}}{(NOA_t + NOA_{t-1})/2} = \frac{\Delta NOA}{\text{avg NOA}}$$

Pg 16 $$\text{Cash-flow-based accruals ratio} = \frac{NI_t - (CFO_t + CFI_t)}{(NOA_t + NOA_{t-1})/2} = \frac{\text{CF based accruals}}{\text{avg NOA}}$$

Learning Module 7 | Financial Statement Modeling

Growth Relative to GDP Growth approach

If company's revenue is forecast to grow at K bps above the nominal GDP growth rate ($g\%$), then company's revenue growth rate = $g\% + \frac{K}{100}\%$

If company's revenue is forecast to grow $H\%$ faster than the nominal GDP growth rate ($g\%$), then company's revenue growth rate = $g\% \times \left(1 + \frac{H}{100}\right)$

Market Growth and Market Share approach

$$\text{Forecast revenue} = \text{Market share (in \%)} \times \text{Industry revenue}$$

Return on Invested Capital

see @ 21 for comp vs RoCE

$$\text{ROIC} = \frac{NOPLAT}{\text{Invested Capital}}$$

where:

$NOPLAT$ = Net operating profit less adjusted taxes

$\text{Invested Capital} = \text{Operating assets} - \text{Operating liabilities}$ = Net Op. Assets

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Paying cash Div to SH who pay no taxes on incm or Div will have NO effect on SH wealth ∴ ΔWealth↑ = cash ↑ ΔWealth↓ = Div (ex-div price of shares).

Noesis

CORPORATE ISSUERS

Learning Module 1 | Analysis of Dividends and Share Repurchases

Dividend Payout Policies

Target payout adjustment model (Lintner model) ~ linear interpolat° to get to Target payout Ratio

$$Expected\ dividend = Last\ dividend + \left(Expected\ Earnings \times Target\ payout\ ratio - Last\ dividend\right) \times Adjustment\ factor$$

where:

$$Adjustment\ factor = \frac{1}{Number\ of\ years\ for\ adjustment\ to\ take\ place}$$

Constant dividend payout ratio policy

$$Dividend = Dividend\ payout\ ratio \times Current\ earnings$$

Video: https://youtu.be/hhcvNiTpZX4 when aft-tax R_d = earnings yld (E/P) ⇒ No chng in EPS aft buyback.

EPS and BVPS After Share Repurchase {Numr has total value} If E/P (ern yld) > $R_d(1-t)$ ⇒ borrow to repurchase → EPS ↑

$$EPS\ after\ buyback = \frac{Earnings\ before\ buyback - After\ tax\ cost\ of\ funds}{Shares\ outstanding\ after\ buyback}$$

Video: https://youtu.be/Pd0-QQF-VhQ If MPx > bvps → bvps ↓ aft repurch; If MPx < bvps, bvps ↑ aft

only inc due to rev. stock split vs spl div, stock div; both reduce bvps.

$$BVPS\ after\ buyback = \frac{Book\ Value\ before\ buyback - Value\ of\ share\ buyback}{Shares\ outstanding\ after\ buyback}$$

Analysis of Dividend Safety

Note: Ret^n of capi. (eg. liqui div) is NOT a taxable event, unlike div or cap. gain.

$$Dividend\ payout\ ratio = \frac{Dividends}{Net\ Income}$$

$$Dividend\ coverage\ ratio = \frac{Net\ Income}{Dividends}$$

$$FCFE\ coverage\ ratio = \frac{FCFE}{Dividends + Share\ repurchases}$$

Div. Imputat^n Tax sys: Eff tax rate paid on cash Div = Personal Tax Rate.

Issuing stock Div does not affect SH wealth.

Greenmail – thwart takeover attempt by paying offering substantial prem. over Mkt px to SH trying to acquire → pay 2go hg

Dutch auction – co. sends an offer to buy sh within a Px range for # sh; SH respond with # sell & Min Px they will accept. Mgmt starts with lowest Px until they get # sh they want to purchase.

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Learning Module 3 | Cost of Capital: Advanced Topics

Weighted average cost of capital

$$WACC = w_d r_d(1-t) + w_p r_p + w_e r_e$$

where:

w_d = Weight of debt in capital structure

w_p = Weight of preferred equity in capital structure

w_e = Weight of common equity in capital structure

r_d = Pre-tax cost of debt

r_p = Cost of preferred equity

r_e = Cost of common equity

Cost of debt, $r_d = r_f + Credit\ spread$

Cost of equity, $r_e = r_f + ERP + IRP$; does not involve β with ERP like CAPM, due to IRP

where:

ERP = Equity risk premium = $Benchmark\ index\ return - Risk\ free\ rate$

IRP = Idiosyncratic risk premium

Leases

$$Present\ Value\ of\ Lease\ Payments + Present\ Value\ of\ Residual\ Value\ to\ Lessor\ (owner) = Fair\ Value\ of\ Leased\ Asset + Lessor's\ Direct\ Initial\ Costs$$

Equity Risk Premium

Historical Approach (Ex-Post)

$$ERP = Average\ benchmark\ index\ return - Average\ risk\ free\ rate$$

Gordon Growth model

$$ERP = \frac{D_1}{V_0} + g - r_f \qquad \because V_0 = \frac{D_1}{r_e - g}$$

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Grinold-Kroner Model

$$ERP = [DY + \Delta(P/E) + i + g + \Delta S] - r_f$$

where:

DY = Dividend yield of market index

$\Delta(P/E)$ = Expected growth rate in P/E

i = Expected inflation = $\frac{1+YTM_{Treasury\ bond}}{1+YTM_{TIPS}} - 1$

g = Expected growth rate in real earnings per share

ΔS = Expected change in shares outstanding

Cost of Equity

Gordon Growth Model

$$r_e = \frac{D_1}{P_0} + g$$

Two-Stage DDM

$$P_0 = \sum_{i=1}^{n} \frac{D_t}{(1+r_e)^t} + \frac{P_n}{(1+r_e)^n}$$

Bond Yield Plus Risk Premium Approach (BYPRP)

$$r_e = r_d + Risk\ premium$$

Capital Asset Pricing Model (CAPM)

$$r_e = r_f + \beta \times ERP$$

Fama-French model

Three-factor model

$$r_e = r_f + \beta_1 ERP + \beta_2 SMB + \beta_3 HML$$

Five-factor model

$$r_e = r_f + \beta_1 ERP + \beta_2 SMB + \beta_3 HML + \beta_4 RMW + \beta_5 CMA$$

where:

SMB = Size premium

HML = Value premium

RMW = Profitability premium

CMA = Investment premium

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Expanded CAPM

$$r_e = r_f + \beta_{peer}(ERP) + SP + IP + SCRP$$

where:

- SP = Size premium (for smaller, privately held companies)
- IP = Industry risk premium
- $SCRP$ = Company-specific risk premium

Build-Up Approach

$$r_e = r_f + ERP + SP + SCRP$$

$$= r_f + ERP + SP$$

Country Spread Model

$$ERP = \text{ERP for a developed market} + \lambda \times \text{Country risk premium (CRP)}$$

where:

λ = Level of exposure of the company in the local country

$Country\ risk\ premium = Sovereign\ yield\ spread$

$$Sovereign\ yield\ spread = \text{Yield on emerging market bonds (denominated in the currency of the developed market)} - \text{Yield on developed market government bonds}$$

Aswath Damodaran's CRP

$$Country\ risk\ premium = Sovereign\ yield\ spread \times \left[\frac{\sigma_{Equity}}{\sigma_{Bond}}\right]_{local}$$

where:

σ_{Equity} = Volatility of the local country's equity market

σ_{Bond} = Volatility of the local country's bond market

International CAPM (ICAPM) – 2-factor

$$E(r_e) = r_f + \beta_G[E(r_{gm}) - r_f] + \beta_C[E(r_C) - r_f]$$

where:

$E(r_{gm}) - r_f$ = Risk premium of a global index

r_C = Wealth-weighted foreign currency index return

Global CAPM (GCAPM): model expected index. ret^n excluding FX risk.

If co's ops are global, but limited to developed countries → can use global CAPM or Intnl CAPM

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Sale & leaseback is a typ of B/S restructuring → realise proceeds from sale of Ⓐ → reduce debt.

Learning Module 4 | Corporate Restructuring

Evaluating Materiality Based on Size

For **Acquisition/Divestiture:**

$$\frac{Value\ of\ transaction}{Enterprise\ value\ of\ acquiring\ company}$$

For **Cost Restructuring:**

$$\frac{Cost\ savings}{Sales}$$

Premium Paid Analysis

$$Takeover\ premium,\ PRM = \frac{DP - SP}{SP}$$ ✓✓

where:

DP = Deal price per share of the target company

SP = Unaffected stock price of the target company (i.e., pre-announcement)

Materiality of acqsn = f(size, fit) → writ strategy & focus

↳ if >10% pre-EV of acquirer

Val calc thr comparable txn Meth faces less scrutiny for mispricing by stkhldrs → litigat° risk ↓

Estimated Tover Px = Est. stock Px from Comps. × (1 + tover prem.)

ESG integration: a) Equity: ID oppt° + ctrl d/s risk; b) Fxd Incm: ctrl D/S risk.

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EQUITY VALUATION

Learning Module 1 | Equity Valuation Applications and Processes

$$V_E - P = (V - P) + (V_E - V)$$

① ②

where:

V_E = Estimated intrinsic value

P = Market price

V = Intrinsic value (actual) ⟶ true but unobsv.

Alpha = ① = Actual mispricing ② = Val° error

Conglomerate discount = Sum-of-the-parts value – Market value

Learning Module 2 | Discounted Dividend Valuation

Discounted Dividend Valuation

$$V_0 = \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t}$$

$$HPR = \frac{P_1 - P_0 + D}{P_0} = \underbrace{\frac{D_1}{P_0}}_{\text{div yld}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{c.g. yld}}$$

Implied c.g. yld = Reqd Ret^n – Div Yld

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \cdots + \frac{D_n}{(1+r)^n} + \frac{P_n}{(1+r)^n}$$

Gordon Growth Model

$$V_0 = \frac{D_1}{r - g} = \frac{D_0(1+g)}{r-g}$$

reqd rate

Fixed-rate perpetual preferred stock

$$V_0 = \frac{D}{r}$$

Value of stock = Value of a company with **zero-growth** + Present value of growth opportunities (PVGO)

perpetual stock; with E_1

$$V_0 = \frac{E_1}{r} + PVGO$$

$$\frac{V_0}{E_1} = \frac{P_0}{E_1} = \frac{1}{r} + \frac{PVGO}{E_1}$$

If dividend and earnings growth rate is constant, ∵ $P_0 = D_1/(r-g)$

$$r = \frac{D_1}{P_0} + g$$

= Div Yld + Exp. Growth; Use combi (tot ret^n) to order potential equity investm when such da

If P/E is –vely related to recent g in Earng but +ve related to anticip future g of E ⇒ cyclical co.

Method of avg RoE → calc norm. EPS (with an adjm in BV, if needed).

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Note: DDM can be used even if co. u'takes share repurchase. → model be accounted for.
Div Dispint of Earnings: LT sus. growth ∝ retn; ∝ $\frac{1}{DPR}$ with const RoE

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Two-Stage Dividend Discount Model

$$V_0 = \sum_{t=1}^{n} \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{D_0(1+g_s)^n(1+g_L)}{(1+r)^n(r-g_L)}$$

Video: https://youtu.be/7vXWsTKiSPE

The H-Model

$$V_0 = \frac{D_0(1+g_L) + D_0H(g_S - g_L)}{r - g_L}$$

where:
H = half-life in years of the high-growth period
g_S = Short-term growth-rate
g_L = Long-term growth rate

Video: https://youtu.be/IAMFZXSPKOY

PRAT model

Sustainable growth rate, $g = b \times ROE$, where b = 1 − DPO ratio = retent^n ratio

Video: https://youtu.be/MnfRRRhuGpA

$$g = \left(\frac{NI - Dividends}{NI}\right) \times \frac{NI}{Sales} \times \frac{Sales}{TA} \times \frac{TA}{TE}$$

Learning Module 3 | Free Cash Flow Valuation

Free Cash Flow to the Firm (FCFF) Valuation Approach

$$Firm\ Value = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1+WACC)^t}$$

If non-operating assets = 0

Equity Value = Firm Value – Market Value of Debt

FCFE Valuation Approach

$$Equity\ Value = \sum_{t=1}^{\infty} \frac{FCFE_t}{(1+r)^t}$$

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Single-Stage (Constant Growth) FCFF and FCFE Model

FCFF Valuation Approach

$$Firm\ Value = \frac{FCFF_1}{WACC - g} = \frac{FCFF_0(1+g)}{WACC - g}$$

FCFE Valuation Approach

$$Equity\ Value = \frac{FCFE_1}{r - g} = \frac{FCFE_0(1+g)}{r - g}$$

Free cash flow to the Firm, FCFF

FCFF = NI + NCC + Int(1 – Tax Rate) – FCInv – WCInv
= CFO + Int(1 – Tax Rate) – FCInv
= EBIT(1 – Tax Rate) + Dep – FCInv – WCInv
= EBITDA(1 – Tax Rate) + Dep(Tax Rate) – FCInv – WCInv

where:
NI = Net income available to common shareholders
NCC = Net noncash charges (e.g. depreciation)
Int = Interest expense
FCInv = Fixed capital investments = Maintenance Capex + Growth Capex
$= \Delta Gross\ PPE = \Delta Net\ PPE + Depreciation$
WCInv = Investment in working capital

Free cash flow to the Equity, FCFE

FCFE = FCFF – Int(1 – Tax Rate) + Net borrowing
= CFO – FCInv + Net borrowing
= NI + NCC – FCInv – WCInv + Net Borrow

where:
Net borrowing = Debt issued – Debt repaid

Video: https://youtu.be/rtlvly6Fl0A

If (FCInv – Dep) and WCInv funded using Debt (based on debt ratio):

FCFE = NI + Dep – FCInv – WCInv + Net borrowing

where:
Net borrowing = DR(FCInv – Dep) + DR(WCInv)

$$DR = Debt\ ratio = \frac{Debt}{Assets}$$

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If company issues preferred shares: // Pref Div added to FCFF ; but not ords Div

FCFF = CFO + Int(1 – Tax Rate) + **Preferred dividends** – FCInv

Two-Stage Free Cash Flow Models

$$Firm\ value = \sum_{t=1}^{n} \frac{FCFF_t}{(1+WACC)^t} + \frac{FCFF_{n+1}}{(WACC-g)}\left[\frac{1}{(1+WACC)^n}\right]$$

$$Equity\ value = \sum_{t=1}^{n} \frac{FCFE_t}{(1+r)^t} + \frac{FCFE_{n+1}}{(r-g)}\left[\frac{1}{(1+r)^n}\right]$$

Value of Firm = Value of operating assets + Value of nonoperating assets
(PV of FCFF)

Learning Module 4 | Market-Based Valuation Price and Enterprise Value Multiples

Enterprise value, EV = Market value of **common stock**
+ Market value of **preferred equity** → also added
+ Market value of **debt** + Minority interest
– Cash and Short-term investments ← but not long-term bw.

$E_0(1-b) = Div_0 \times (1+g) = Div_1$

$g = b \times RoE$ &

$RoE = \frac{EPS}{Avg\ E_q\ /sh}$

	Actual	Justified
Trailing P/E	$\frac{Market\ price\ per\ share}{EPS\ over\ previous\ 12\ months}$	$\frac{(1-b)(1+g)}{r-g}$
Leading P/E	$\frac{Market\ price\ per\ share}{Forecasted\ EPS\ over\ next\ 12\ months}$	$\frac{1-b}{r-g}$
P/B	$\frac{Market\ price\ per\ share}{Book\ value\ per\ share}$	$\frac{ROE-g}{r-g}$
P/S	$\frac{Market\ price\ per\ share}{Sales\ per\ share}$	$\frac{V_0}{S_0} = \frac{E_0}{S_0} \times \frac{(1-b)(1+g)}{r-g}$ or $\frac{V_1}{S_1} = \frac{E_1}{S_1} \times \frac{1-b}{r-g}$

Rem: to adjust for non-recurring earnings OR for discontinued ops.

No need to grow.

Video: https://youtu.be/c0vmCUtDpZs

forecasted ←

$RoIC = \frac{NOPLAT}{Invest\ Cap} = \frac{NOPAT - Adj\ Taxes}{Op\ (A) - Op\ (L)}$; better than RoE ∵ i) unaffected by deg of fin. leverg. Higher RoIC – exploit competitive advtg

$RoCE = \frac{Op.\ \Pi}{Cap.\ Emp.} = \frac{Op.\ Profits}{D+E}$ ≃ RoIC before taxes ; ∴ compare companies w. diff. tax rates.

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	Actual	Justified
Trailing D/P	$\frac{4 \times Most\ recent\ quarterly\ dividend}{Market\ price\ per\ share}$	$\frac{r-g}{1+g}$
Leading D/P	$\frac{Forecast\ dividends\ over\ the\ next\ year}{Market\ price\ per\ share}$	$r-g$
Earnings yield	$\frac{EPS}{Price\ per\ share}$	$\frac{r-g}{(1-b)(1+g)}$

Underlying Earnings = EPS – non recurring gains + non recurring loss

Normalized Earnings

Method 1: Average EPS Approach

$$\text{Normalized EPS} = \frac{1}{n}\sum_{i=1}^{n} EPS_i$$

Method 2: Average ROE Approach

$$\text{Normalized EPS} = \frac{1}{n}\sum_{i=1}^{n} ROE_i \times Current\ Book\ value\ per\ share$$

Price-to-Earnings Growth (PEG) Ratio

$$PEG\ ratio = \frac{P/E\ ratio}{g\ (in\ \%)}$$

Momentum Indicators

Earnings surprise = Reported EPS – Expected EPS

$$\textbf{Scaled earnings surprise} = \frac{Reported\ EPS - Expected\ EPS}{\sigma(Analyst\ forecast\ EPS)}$$

$$\textbf{Standardized unexpected earnings (SUE)} = \frac{Earnings\ Surprise}{\sigma(Earnings\ Surprise)}$$

Portfolio P/E

$$Weighted\ harmonic\ mean = \frac{1}{\sum_{i=1}^{n} \frac{w_i}{X_i}}$$

where:

w_i = Weight of stock *i* in portfolio

X_i = P/E of stock *i*

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RI = (RoIC − Eff. capi chrg) × Tot cap. ; where Tot cap = D+E ; Eff cap chrg = $\frac{1}{D+E}(R_eE + R_d(1-t)D)$

RI = NI − eqty charge ; eqty chrg = $BV_{t-1} \times R_e$

Noesis

Learning Module 5 | Residual Income Valuation

✓ **Economic Value Added (EVA)** → lagged

$$EVA_t = EBIT_t(1-T) - (WACC \times Invested\ Capital_{t-1})$$

Market Value Added (MVA)

$$MVA_t = Market\ value\ of\ Firm_t - Invested\ Capital_t$$

✓ **Residual Income, RI** → read rate = cost of equity

$$RI_t = E_t - (r \times B_{t-1}) = (ROE - r) \times B_{t-1} \quad \because RoE = \frac{NI}{BV_{t-1}E} = \frac{EPS}{bvps}$$

Residual Income Model

$$V_0 = B_0 + \left[\frac{RI_1}{(1+r)^1} + \frac{RI_2}{(1+r)^2} + \frac{RI_3}{(1+r)^3} + \cdots\right]$$

Video: https://youtu.be/O0KTBkEtP9M

✓ **Single-stage residual income valuation model**

$$V_0 = B_0 + \frac{(ROE - r) \times B_0}{r - g} = B_0 + \frac{RI_1}{r - g}$$; builds on top of B_0

Video: https://youtu.be/82GJu5umrB0

✓ **Tobin's Q**

$$Tobin's\ Q = \frac{Market\ value\ of\ debt + Market\ value\ of\ equity}{Replacement\ cost\ of\ total\ assets}$$

Continuing Residual Income → Hi Persistence → mkt leader position // low expected rate of decay → low div ; hold back cash for reinvmt

$$V_0 = B_0 + \sum_{t=1}^{T-1} \frac{RI_t}{(1+r)^t} + \frac{RI_T}{(1+r-\omega)(1+r)^{T-1}} \qquad 0 \le \omega \le 1$$

ω = Persistence factor ; Low if low level of accruals in I/S.

If RI declines to **Long-run level** in mature industry, with premium over book value → discount from final T

$$V_0 = B_0 + \sum_{t=1}^{T} \frac{RI_t}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

Video: https://youtu.be/vhRW3q70E0w

✓ Clean surplus relationship:

$$B_t = B_{t-1} + E_t - Div_t$$

DDM & Resi Incm models shld provide same results, provided assmpns are consistent b/c similar. For items that bypass the I/S, may have to adjust the NI before Resi Incm Val^n. Eg. add OCI to NI.

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Income app:- use PV; DCF // cos with predictable c/f [Inc. Val]
Mkt app. - when comparable public firms exist [Rel. Val.]
Asset-based :- (A) (L) ; diff to pred c/f; unique predtt. → No [Abs. Val] comps.

Noesis

Learning Module 6 | Private Company Valuation

Capitalized Cash Flow Method (CCM)

$$Firm\ value = \frac{FCFF_0(1+g)}{WACC - g} \rightarrow \begin{matrix} Equity \\ value \end{matrix} = \begin{matrix} Firm \\ value \end{matrix} - \begin{matrix} Market\ value \\ of\ Debt \end{matrix}$$

$$Equity\ value = \frac{FCFE_0(1+g)}{r - g}$$

Entp Val (EV) = MV(D) + MV(E) − Cash & Mktbl Secu
(= L.T. Invm)

Excess Earnings Method (EEM)

$$\begin{matrix} \textbf{Excess} \\ \textbf{earnings} \end{matrix} = \begin{matrix} Normalized \\ earnings \end{matrix} - \begin{matrix} Earnings\ required\ to\ provide \\ the\ required\ rate\ of\ return\ on \\ \textbf{working capital and fixed assets} \end{matrix}$$

$$Value\ of\ the\ intangible\ assets = \frac{(Excess\ Earnings)_1}{k - g}$$

Value of the firm = Working capital + Fixed assets + Intangible Assets

MVIC by EBITDA ratio for pvt cos, where MVIC = mkt val of invested capital. < for pvt. cos. >
Video: https://youtu.be/137ga1xgAbA
With ctrl prem = MVIC × (1 + C.P.)

Control Premium

$$\begin{matrix} Equity\ value \\ (with\ control\ premium) \end{matrix} = \begin{matrix} Equity\ value \\ (without\ control\ premium) \end{matrix} \times (1 + Control\ premium)$$

further inc based on D/A ← $Adjusted\ control\ premium = Control\ premium \times \left(1 + \frac{Debt}{Assets}\right)$

Discount for Lack of Control and Marketability
(→ DLOC) (→ DLOM)

Discount for Lack of Control (DLOC)

$$DLOC = 1 - \frac{1}{1 + Control\ premium}$$

✓ Total discount = 1 – (1 – DLOC)(1 – DLOM) ; the discounts are multiplicative

✓ $DLOM = \frac{Value\ of\ ATM\ option}{Share\ price}$

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Vol of ST rates is higher than of LT rates & linked to monetary policy, vol of LT rates are [illegible]
Pref. Habitat Th :- If expected addnl ret^n to given becomes large enuf, instl^ deviate from their preferred habitats or matu

Noesis

FIXED INCOME

Learning Module 1 | The Term Structure and Interest Rate Dynamics

Forward Pricing Model

$$DF_B = DF_A \times F_{A,B-A}$$

where:

$$DF_B = \frac{1}{(1+z_B)^B}$$

$$F_{A,B-A} = \frac{1}{(1+f_{A,B-A})^{B-A}}$$

inverse of

Forward Rate Model

$$(1+z_B)^B = (1+z_A)^A(1+f_{A,B-A})^{B-A}$$

where:

z_B = Spot rate for period B

$f_{A,B-A}$ = $(B-A)$ forward rate that starts in period A

Same as FRA, 2.1

Calculating spot rate from one-period forward rates

$$z_T = \left[(1+z_1)(1+f_{1,1})(1+f_{2,1})\dots(1+f_{T-1,1})\right]^{1/T} - 1$$

Boostrapping Spot Rates From Par Rates

Video: https://youtu.be/-FnweFO172Q

Fixed swap rate

$$s_T = \frac{1-DF_T}{\sum_{t=1}^{T} DF_t} = \frac{1 - Discount\ Factor\ of\ Last\ Payment}{Sum\ of\ Discount\ Factors}$$

Swap spread = YTM of swap rate – YTM of government bond (same maturity) → corresp. to bond under review → may have to interpolate

TED spread = LIBOR – YTM of T-bill (same maturity)

LIBOR-OIS spread = LIBOR – OIS Fixed rate

For **Parallel shifts** in yield curve:

Recm

$$\%\Delta PV = -ModDur \times \Delta YTM \qquad \Delta PV = -ModDur \times \Delta YTM \times PV_0$$

$$\%\Delta PV = -EffDur \times \Delta Curve \qquad \Delta PV = -EffDur \times \Delta Curve \times PV_0$$

@ Pg 32

For ZCB, duration = Maturity; In Pfo Eff Durn → Recm. to use MV of for wtg, not FV.

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Key Rate Durn: meas pfo risk exposr to each key rate. Σ KRd$_i$ = Eff krd for pfo.

Non-parallel shifts (i.e. change in slope or curvature):

$$\%\Delta PV = -KeyRateDuration \times \Delta Key\ Rate$$

Learning Module 2 | The Arbitrage-Free Valuation Framework

Arbitrage-free Value of Bond

$$V_0 = \frac{C}{(1+z_1)^1} + \frac{C}{(1+z_2)^2} + \cdots + \frac{FV+C}{(1+z_n)^n}$$

where:
z_n = Spot rate for period n i.e. discount each cff by separate spot rate to present.

Backward Induction Valuation Methodology (keep folding blwrd) Rem to add

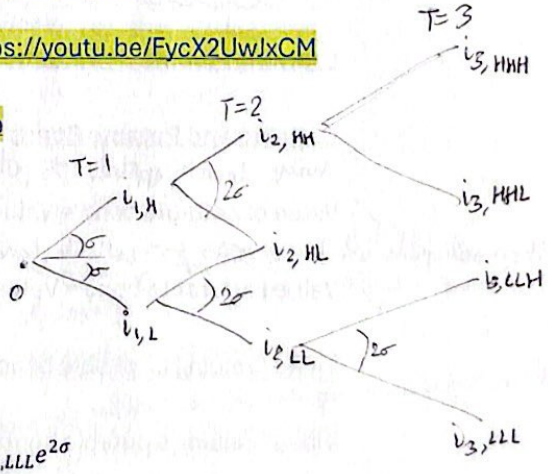
$$Bond\ value\ at\ any\ node = \frac{(0.5 \times V_H + 0.5 \times V_L) + C}{1+i}$$

where:
V_H = bond's value if the higher forward rate is realized one year hence
V_L = bond's value if the lower forward rate is realized one year hence
C = coupon payment that is not dependent on interest rates

Video (Backward Induction Valuation): https://youtu.be/DhAVQ3hlXlQ

Video (Backward Induction with Financial Calculator): https://youtu.be/FycX2UwJxCM

Video (Pathwise Valuation): https://youtu.be/3oM-220oi7o



Binomial Interest Rate Tree

$i_{1,H} = i_{1,L}e^{2\sigma}$

$i_{2,HH} = i_{2,LL}e^{4\sigma}$ $\quad i_{2,HL} = i_{2,LL}e^{2\sigma}$

$i_{3,HHH} = i_{3,LLL}e^{6\sigma}$ $\quad i_{3,HHL} = i_{3,LLL}e^{4\sigma}$ $\quad i_{3,LLH} = i_{3,LLL}e^{2\sigma}$

Higher int rate vol ⇒ Val (c)↑ & Val (p)↑ options ; $V_{putb} = V_{bond} + V_{put}$; ∴ σ↑ ⇒ V_{putb}↑ ⇒ a higher yld & OAS↑

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Equilibrium Term Structure Models

Cox-Ingersoll-Ross (CIR) Model

$$dr = k(\theta - r_t)dt + \sigma\sqrt{r_t}dz$$

; Vol = $f(\sqrt{r_t}) \neq$ const

Vasicek Model

$$dr = k(\theta - r_t)dt + \sigma dz$$

; Vol is const

where:

- k = Speed of reversion (> 0)
- θ = Long-run interest rate
- σ = Interest rate volatility

Arbitrage Free Models

Ho-Lee Model

$$dr_t = \theta_t dt + \sigma dz_t$$

→ can be modeled w. const vol & calib to match y/c

Kalotay-Williams-Fabozzi (KWF) Model

$$d(\ln r_t) = \theta_t dt + \sigma dz_t$$

where:

θ_t = Time-dependent drift term

Learning Module 3 | Valuation and Analysis - Bonds with Embedded Options

Callable and Putable Bonds

When y/c flattens, val of call opt^n ↑ but V(put opt) ↓ ∵ investors have lower opptnts to put back the bond.

- Value of **callable** bond = Value of straight bond – Value of **issuer call option**
- Value of **putable** bond = Value of straight bond + Value of **investor put option**

When int rate vol ↓ → OAS for callable bonds increases → relative cheapness increases. Val ↑ but OAS also ↑

Video (Valuing a callable bond): https://youtu.be/lWLSodiqZaM

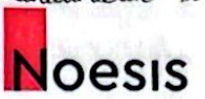
Video (Valuing a putable bond): https://youtu.be/qmUnAtpXIAg

$$Effective\ duration = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Curve) \times PV_0}$$

$$Effective\ convexity = \frac{(PV_-) + (PV_+) - 2 \times PV_0}{PV_0 \times (\Delta Curve)^2}$$

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Conv. Bonds may have covenants that allow investors to put bond back to Issuer, convert @ lower 'chng of ctrl' conv. px, or both. This is especially when firm gets acquired & acquirer is unattractive to investors.



Capped and Floored Floaters

C/C match ↔ Value of capped floater = Value of straight floater – Value of cap (minus like the callable bond)

Value of floored floater = Value of straight floater + Value of floor

Video (Valuing a capped floater): https://youtu.be/d4LNMdXV9vU

Video (Valuing a floored floater): https://youtu.be/YJZU0THHBNE

Convertible Bonds

Issuers may call a convtbl bond when u/l sh px > conv. px to "force" investors to convert & ∴ reduce eliminate further coupon payout.

Valued using Arb free binomial Int Rate Tree

$$\text{Conversion value} = \text{Underlying share price} \times \text{Conversion ratio}$$

→ OR Conv. Ratio = Par Val of bond / Curr. conv. Px

$$\text{Minimum value of convertible bond} = \text{Max}\left[\text{Conversion value}, \text{Value of underlying Straight bond}\right]$$

$$\text{Market conversion price} = \frac{\text{Convertible bond price}}{\text{Conversion ratio}}$$

OR Conv. bond Px = Mkt Conv Px × Conv. Ratio

$$\text{Market conversion premium per share} = \text{Market conversion price} - \text{Underlying share price}$$

$$\text{Market conversion premium ratio} = \frac{\text{Market conversion premium per share}}{\text{Underlying share price}}$$

$$\text{Premium over Straight value} = \frac{\text{Convertible bond price}}{\text{Straight value}} - 1$$

Convertible Bond (With No Additional Options)

$$\text{Value of convertible bond} = \text{Value of straight bond} + \text{Value of call option on issuer's stock}$$

MinVal of conv. B = Max [Conversion val of conv B ; option free bond]

→ subtract val of call option, as before, on 32

Callable Convertible Bond

$$\text{Value of convertible bond} = \text{Value of straight bond} + \text{Value of call option on issuer's stock} - \text{Value of issuer call option}$$

$V_{conv} = V_{st} + V(\text{option exer by Investor}) - V(\text{option exer by Issuer})$

Putable Convertible Bond

$$\text{Value of convertible bond} = \text{Value of straight bond} + \text{Value of call option on issuer's stock} + \text{Value of investor put option}$$

When Issuer's share Px < Bond conv. Px, it resembles ↔ issuer's common stock from risk-ret" persp.

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VND = Σ(diff × disc. fac.) i.e. Val w/ No Def.

Learning Module 4 | Credit Analysis Models

$$G\text{-}spread = YTM\ of\ Corporate\ bond - YTM\ of\ Government\ bond$$ → measure of cred risk for c-bonds

$$Loss\ given\ default = Expected\ exposure \times \left(1 - Recovery\ rate\right)$$

$$Loss\ severity = 1 - Recovery\ rate$$ → already includes EAD

Expected Loss = Probability of Default × Loss Given Default → used to evaluate credit risk. ∝ c.Risk

$$Fair\ value\ of\ credit\ risky\ bond = Fair\ value\ of\ bond\ assuming\ no\ default - Credit\ Valuation\ Adjustment$$

$$Credit\ valuation\ adjustment,\quad CVA = \sum_{t=1}^{n} \frac{EL_t}{(1+rf_t)^t} = \sum_{t=1}^{n} \frac{POD_t \times LGD_t}{(1+rf_t)^t}$$

where:

EL_t = Expected loss of bond at time t

POD_t = Probability of default of bond at time t

LGD_t = Loss given default at time $t = Expected\ Exposure_t - Recovery_t$

rf_t = Risk-free rate at time t

n = Bond's remaining tenor

$$PV\ of\ expected\ loss\ for\ period\ t = \frac{EL_t}{(1+rf_t)^t}$$

$$POD_t = (1 - Hazard\ rate)^{t-1} \times Hazard\ rate$$ POS = prob. of survival

Approximation of credit spread ≈ Annual hazard rate × (1 – Recovery rate) Note

Video (Probability of Default): https://youtu.be/e7K4x48Eg4U

Video (Valuing a Credit Risky Bond – Zero Interest Rate Volatility): https://youtu.be/2l9bgu-o7aI

Video (YTM of Corporate Bonds – Default and Non-Default): https://youtu.be/K253Y7c2Yto

Expected percentage price change of a corporate bond due to ΔRatings

$$\sum Probability\ of\ credit\ migration \times \%\Delta P$$

// & transition evaluated across all rating Δs.

If ques asks for pxc. change in bond due to migrat", ignore coupon (v/s Δtot ret"-cpn) (From/To ≡ Col/Row)

where:

$\%\Delta P = -ModDur \times \Delta credit\ spread$ = diff b/w existing spread v/s post-rating-chng spread.

Med" term, granular loans, Homogen → pfo approach
Short " " " " → stats " (∵ pfo is ≃ static)
Med " Non-gran " Hetero → Loan-by-Loan

Expected ret" over next year = YTM + %ΔP (due to rating)

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Starting a new loan will help to inc credit score. // Credit Transit Matrix does not capture the possib of defaul
Default rating = f(PD) ; Credit Rating = f(PD, LGD); Prob of rating Δ is asymm around existing rating; Spread ↑
for D'grade > U'grade Risk-neut PD > hist to def rate bec rating due to uncertainty over timing of def, liqui, taxeat

Noesis

Structural Model : A1) Ⓐ trade in frictionless, no-arb mkt; A2) Liab are equiv to ZCB.
↳ analyze B/S & estim def barriers based on Liab.; can explain eco reasons for def unlike reduced form.

$$A_t = D(t,T) + S_t$$

In terms of...	Call options	Put options
Equity	$E(T) = Max[A(T) - K, 0]$	$E(T) = A(T) - K + Max[K - A(T), 0]$
Debt	$D(T) = A(T) - Max[A(T) - K, 0]$	$D(T) = K - Max[K - A(T), 0]$

↳ owning co. Ⓐ & having a short posn in a call option on co. Ⓐ. If V(Ⓐ) < V(B) → No call
from E(T)
↳ K = fV(D)

where:
S_t = Equity value at time t
A_T = Asset value at time T
K = Face value of debt

Learning Module 5 | Credit Default Swaps In CTD, may not match for maturity.; check pari-pas or higher ranking.

CDS payout amount = Payout ratio × Notional
= (1 – Recovery rate of CTD bond) × Notional
↳ respect to Deliver

Upfront payment = PV of protection leg – PV of premium leg

= bucket price of CDS ✓

$$\text{Upfront premium} = \text{PV of Credit Spread} - \text{PV of Fixed Coupon}$$

↳ if >0 ⇒ buyer pays at T=0 & vice-versa

$$\approx \left(\text{Credit Spread} - \text{Fixed Coupon}\right) \times \text{CDS Duration}$$ ✓; all terms from CDS

✓ $$\text{Price of CDS per 100 notional} = 100 - \text{Upfront premium}$$; if prem < 0 ⇒ price of CDS > 100

$$\%\ \text{Change in CDS price} = \text{Change in spread in bps} \times \text{Duration}$$

For HY bonds, stated fixed rate in CDS = 5%

As credit spread widen, cred risk ↑ ⇒ Upfront prem ↑

If spd^e ↑ ⇒ BondVal ↓ ; bet short the credit qual. & vice-versa.

D/fly for Treas Bond fut contr is a generic bond → may have diff cpn than the bond deliv'd to settle

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DERIVATIVES

Learning Module 1 | Pricing and Valuation of Forward Commitments

Forward Contracts

Forward Pricing:

$F_0 = S_0(1+r)^T$ ⇒ $F_t = S_t(1+r)^{T-t}$

$F_0 = (S_0 + CC_0 - CB_0)(1+r)^T$ (cost, benefit)

$F_0 = S_0 e^{r_c T}$

$F_0 = S_0 e^{(r_c + CC - CB)T}$ → all rates continuously comp.

where:

S_0 = Current spot price

F_0 = Forward price (set today)

r = Annually compounded risk-free rate

r_c = Continuously compounded risk-free rate

CC_0 = PV of Carry cost

CB_0 = PV of carry benefits

CC = Continuously compounded cost of carry

CB = Continuously compounded carry benefit → eg. div yld

Forward Valuation (Long Position):

$V_0 = 0$

$\because F_t = S_t(1+r)^{T-t}$

any time, t $V_t = \frac{F_t - F_0}{(1+r)^{T-t}} = S_t - \frac{F_0}{(1+r)^{T-t}}$

At Maturity, T $V_T = S_T - F_0$

~~YTM~~ is poor estimate of exp. ret" if 1) int rate volatile, 2) yc up/dn sloping, 3) of default risk, 4) Options ✓

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Forward Rate Agreement (FRA)

$$\textit{Long FRA payoff at expiration of FRA} = \frac{Notional[L_m - FRA_0]t_m}{1 + D_m t_m}$$

$$FRA_0 = \left(\frac{1 + L_T t_T}{1 + L_h t_h} - 1\right)\left(\frac{1}{t_m}\right)$$

Valuation at time $t = g$ (prior to FRA expiration):

$$\textit{Value of Long FRA at } g = \frac{Notional(FRA_g - FRA_0)t_m}{1 + D_{T-g} t_{T-g}}$$

where:

D_m = Discount rate for m periods at $t = h$

h = FRA tenor

m = Tenor of the underlying rate

$T = h + m$ = Maturity of underlying instrument

If sold FRA (floating) to recv Fixed rate, <adv. settld; FRA

settlmt payont $= \frac{\text{Ntl}^n\ (\text{fx} - \text{fl. rate}) \times 180/360}{\left(1 + \text{fl. rate} \times \frac{180}{360}\right)^T}$

Video (Pricing an FRA): https://youtu.be/uBmAt_z9f3Y

Video (Valuing an FRA): https://youtu.be/AYKRVdaYvxY

Fixed Income Forwards and Futures

$$VF_t = \frac{\left[(S_t + AI) - PV(Incm)\right] \times (1+r)^T - AI}{CF}$$

(T: time to maturity of bond)

S_t = current Px ; AI = accrued int.

PV(Incm) ≡ from upcoming cpn; discounted st

Pricing:

$$F_0 = \underset{price}{Quoted\ futures} \times \underset{factor\ (CF)}{Conversion}$$
$$= FV(B_0 + AI_0) - AI_T - FVCI$$

Valuation for fixed income **forward contracts**:

$V_t = Present\ value\ of\ difference\ in\ forward\ prices$
$= PV[F_t - F_0]$

Valuation for fixed income **futures contracts**:

V_t = Price change since previous day's settlement

where:

B_0 = Quoted bond price

$$AI = \frac{Number\ of\ accrued\ days\ since\ last\ coupon\ payment}{Total\ days\ during\ the\ coupon\ payment\ period} \times \frac{Annual\ coupon}{Coupon\ frequency}$$

pro-rata cpn for accrued days

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Interest Rate Swaps (IRS)

$$FS = \frac{1 - PV_n}{\sum_{i=1}^{n} PV_i}$$

$$PV_i = \frac{1}{1 + Spot\ rate_i\left(\frac{Days\ to\ Maturity_i}{360}\right)} = DF_i$$

Pay-fixed, receive-floating IRS — (A)

rcv. float → pay fixed

$$Value\ of\ Swap = Notional \times (FS_t - FS_0) \sum_{i=1}^{n} PV_i \quad — (A)$$

Receive-fixed, pay-floating IRS → opp. of (A)

$$Value\ of\ Swap = Notional \times (FS_0 - FS_t) \sum_{i=1}^{n} PV_i$$

Video (Pricing an Interest Rate Swap) : https://youtu.be/0QvtKZutr5E

Video (Valuing an Interest Rate Swap): https://youtu.be/_A2a909etvg

Currency Swap → greater c/p risk exposure @ swap expiry, v/s IRS bk ccy swap exchng notional @ beg'n/g & end of swap. ∴ Periodic payments in diff ccy & are paid separately; not netted.

IRS

Pricing for fixed leg of currency swap in currency a

$$FS_a = \frac{1 - PV_{n,a}}{\sum_{i=1}^{n} PV_{i,a}}$$

Value of a fixed-for-fixed currency swap

$$V_{CS} = Notional_a \times V_a - S_t \times Notional_b \times V_b$$

recv. pay

$$V_a = FS_a \sum_{i=1}^{n} PV_{i,a} + PV_{n,a} \times Par_a = Value\ of\ currency\ a\ leg\ (receive)$$ addnl in ccy v/s IRS

$$V_b = FS_b \sum_{i=1}^{n} PV_{i,b} + PV_{n,b} \times Par_b = Value\ of\ currency\ b\ leg\ (pay)$$

S_t = Spot exchange rate at time t (quoted as a/b)

Video (Pricing a currency swap): https://youtu.be/XZlxcVByc00

Video (Valuing a currency swap): https://youtu.be/3h4mElS48aA

Ccy swap: initial ccy exch of b/w parties @ $t=0$. Each period int is based on ccy submitted (notional) x r x time

OIS: type of IRS where swap's periodic floating rate = GM (daily unsecured o/n rate b/w banks)

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Equity Swap

Value of equity swap (receive fixed-rate, pay equity return)

$$V_{EQ,t} = V_{FIX}(C_0) - \frac{S_t}{S_{t-1}} \times Notional - PV_t(Par - Notional)$$

$$Value\ of\ Equity\ Leg = \frac{S_t}{S_{t-1}} \times Notional$$

Cash flow for equity leg = $Notional \times Periodic\ equity\ return$

where:
$V_{FIX}(C_0)$ = Value at time t of a fixed-rate bond initiated with coupon C_0 at Time 0
S_t = Current equity index level
S_{t-1} = Equity index level at last reset date

Learning Module 2 | Valuation of Contingent Claims

Hedge Ratio

$$h_{call} = \frac{c^+ - c^-}{S^+ - S^-} \geq 0$$

$$h_{put} = \frac{p^+ - p^-}{S^+ - S^-} \leq 0$$

No-arbitrage Approach:

$$c = h_{call}S + PV(-h_{call}S^+ + c^+) = h_{call}S + PV(-h_{call}S^- + c^-)$$

$$p = h_{put}S + PV(-h_{put}S^+ + p^+) = h_{put}S + PV(-h_{put}S^- + p^-)$$

Expectations Approach:

$$\pi = \frac{(1+r) - d}{u - d}$$

where:
u = Up factor
d = Down factor
r = Risk-free rate

One-period Binomial Model → simple expected value & 1-time discounting

$$c = \frac{\pi c^{+} + (1-\pi)c^{-}}{1+r}$$

$$p = \frac{\pi p^{+} + (1-\pi)p^{-}}{1+r}$$

where:
π = Risk-neutral probability of an up-move

Rem

Note: For **interest rate options**, $\pi = 0.5$ and discount expected option payoff using the 1-period forward rates.

Video (Valuing interest rate options): https://youtu.be/X4R8j_cf8SA

helpful

Two-period Binomial Model:

$$c = \frac{\pi^2 c^{++} + 2\pi(1-\pi)c^{+-} + (1-\pi)^2 c^{--}}{(1+r)^2}$$; formula ~ $(a+b)^2$ symmetry → discounting over 2 periods ∴ "square"

$$p = \frac{\pi^2 p^{++} + 2\pi(1-\pi)p^{+-} + (1-\pi)^2 p^{--}}{(1+r)^2}$$

F For **2-period American-styled call option** with dividend in $t = 1$: → remove benefit ; add costs

$$S^{+} = u \times (S - PV\ of\ dividends\ at\ risk\ free\ rate)$$

$$S^{-} = d \times (S - PV\ of\ dividends\ at\ risk\ free\ rate)$$

Video: https://youtu.be/U_XkIZjJIAU

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BSM assumptions : 1) stock Pxs are log-norm ; 2) $LN\left(\frac{S_t}{S_0}\right) \sim N(0,1)$ i.e. nat-log of rets are normal ; 3) Kg of σ (vol) assumed constant.

Black-Scholes Option Pricing Model

leveraged posn in ully stocks, s.t. $c = SN(d_1) - Xe^{-rT}N(d_2)$ $N(d_1)$ shares bot by short selling $N(d_2)$ ZCB

$$p = Xe^{-rT}N(-d_2) - SN(-d_1)$$

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Put-call parity: $p + S = c + Xe^{-rT}$

- Hedge ratio for calls $= N(d_1)$
- Probability that the call option expires in the money $= N(d_2) = Prob(S_T > X)$
- Hedge ratio for puts $= N(d_1) - 1 = -N(-d_1)$
- Probability that the put option expires in the money $= 1 - N(d_2)$

$$Prob(S_T < X) = N(-d_2)$$

BSM model with carry benefits → replace S with $Se^{-\gamma T}$

$$c = Se^{-\gamma T}N(d_1) - Xe^{-rT}N(d_2)$$

$$p = Xe^{-rT}N(-d_2) - Se^{-\gamma T}N(-d_1)$$

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r - \gamma + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

$d_2 = d_1 - \sigma\sqrt{T}$ → same as before

Put-call parity: $p + Se^{-\gamma T} = c + Xe^{-rT}$; only modify the S component

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IR floor = series of IR put options ; IR cap = series of IR call options

Black Option Valuation Model

European Options on Futures

$$c = e^{-rT}[F_0(T)\,N(d_1) - XN(d_2)]$$

$$p = Xe^{-rT}N(-d_2) - Se^{-rT}N(-d_1)$$

$$d_1 = \frac{\ln\left[\frac{F_0(T)}{X}\right] + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Put-call parity:

$$c = e^{-rT}[F_0(T) - X] + p$$

Interest Rate Options

$$c = (AP)e^{-r(t_{j-1}+t_m)}\left[FRA(0, t_{j-1}, t_m)N(d_1) - R_X e^{-rT}N(d_2)\right]$$

$$p = (AP)e^{-r(t_{j-1}+t_m)}\left[R_X e^{-rT}N(-d_2) - FRA(0, t_{j-1}, t_m)N(-d_1)\right]$$

$$d_1 = \frac{\ln\left[\frac{FRA(0, t_{j-1}, t_m)}{X}\right] + \left(\frac{1}{2}\sigma^2\right)t_{j-1}}{\sigma\sqrt{t_{j-1}}}$$

$$d_2 = d_1 - \sigma\sqrt{t_{j-1}}$$

Payer Swaption

$$PAY_{SWN} = AP \times [R_{FIX}N(d_1) - R_X N(d_2)] \times \sum_{j=1}^{n} PV_j(1)$$

Receiver Swaption

$$REC_{SWN} = AP \times [R_X N(-d_2) - R_{FIX}N(-d_1)] \times \sum_{j=1}^{n} PV_j(1)$$

$$d_1 = \frac{\ln\left(\frac{R_{FIX}}{R_X}\right) + \left(\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

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Video (Interest Rate Options & Swaptions Equivalences: https://youtu.be/uZQO50sEzso

Optimal Number of Hedging Units (for Delta Hedging) → neutralize exposure to dlying px changes. But delta w/estm the impact for large ↑ or ↓

$$N_H = -\frac{Portfolio\ delta}{Delta_H}$$

Video: https://youtu.be/v8RcvkQKFpw

Option Greeks

$Delta_{call} = e^{-\delta T} N(d_1)$

$Delta_{put} = -e^{-\delta T} N(-d_1)$

where:

δ = Continuously compounded dividend yield

$\Theta = \frac{\partial c}{\partial t}$; More time to expiry, higher val of call & puts.

$\rho = \frac{\partial opt^n}{\partial i}$; +ve related for call; benefit of not having hi i
-ve related for put; higher inv'm incm foregone

$Vega = \frac{\partial opt}{\partial \sigma}$; +ve related for calls & puts.

$Gamma_{call} = Gamma_{put} = \frac{e^{-\delta T}}{S\sigma\sqrt{T}} N(d_1)$ ≈ $\frac{Delta}{S\sigma\sqrt{T}}$

$c = c_0 + Delta_{call} \times \Delta S + \frac{1}{2} Gamma_{call} \times (\Delta S)^2$ ~ Dur'n + convexity effects

$p = p_0 + Delta_{put} \times \Delta S + \frac{1}{2} Gamma_{put} \times (\Delta S)^2$

A short straddle maybe used when lower vol. is expected.

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Value creation in LBO include: 1) debt red", 2) earnings growth, 3) Multiple expansn.
Excess ret" swap: – short party pays when index is over exercise Px.

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ALTERNATIVE INVESTMENTS

Learning Module 1 | Introduction to Commodities and Commodity Derivatives

$$\text{Futures price} = \text{Spot price of physical commodity} + \text{Storage costs} - \text{Convenience yield}$$

$$\text{Calendar spread} = \text{Near term futures contract closing price} - \text{Longer term futures contract closing price}$$

⇒ Contango ⇒ –ve
Bkwdn ⇒ +ve

$$\text{Price return} = \frac{\text{Current price} - \text{Previous price}}{\text{Previous price}}$$

same logic

$$\text{Roll return} = \frac{\left(\text{Near term futures contract closing price} - \text{Longer term futures contract closing price}\right)}{\text{Near term futures contract closing price}} \times \text{Percentage of the position in the futures contract being rolled}$$

$$\text{Total return} = \text{Price return} + \text{Roll return} + \text{Collateral return} + \text{Rebalancing return (for index only)}$$

↳ not for index contracts
↳ yld earned by funds dep on margin to maintain fut posn → proxy = Rf 2 but also adj for fract" of Yr.

Insurance Rsp: Producers need' cover → sale of fut → fut Px ↓ ⇒ Backwdn" ⇒ LT fut Px < S.T. fut Px / contango
Hedging Pressure Rsp: 3 scenarios blw Producers & Consumers & ∴ lead to Bkwdn, Contango or flat.
Storage Th: Conv. Yld ∝ 1/Inv.Supp & ∝ 1/Availability → Contango w. fut delivery px > spot. ; Hi Inv ⇒ storg cost > conv Yld
Speculators benefit when futures price deviates from spot px. → negative roll ret"
Producers are sellers → Hedging down dominates → Bkwdn"

Learning Module 2 | Overview of Types of Real Estate Investment

Net and Gross Leases

$$\text{Net rent} = \text{Gross rent} - \text{Operating expenses}$$

Retail Rent

$$\text{Rent per square foot} = \text{Minimum rent per square foot} + \text{Share\%} \times \left(\text{Revenue per square foot} - \text{Natural breakpoint per square foot}\right)$$

Appraisal-based index

cap. gain

$$\text{Return} = \frac{\text{NOI} - \text{Capital Expenditures} + \left(\text{Ending market value} - \text{Beginning market value}\right)}{\text{Beginning market value}}$$

REoc: less regulated than REIT; ∴ various leverage structrs → degrees of fin. leverag.; REOC taxed like ordi co.
↳ lower mkt val" ∵ equity mkts prefer the tax advtg of REIT & hi incm distrib" (REIT)
↳ 1) rental prop ≥ 75% of REIT Ⓐ 2) ~90% of REIT taxbl incm be distbd as Div → ∴, little devlp of prop
Hence REOC

Altho EBITDA is non-Gaap, it is accepted by SEC for reporting REIT pfmnc

REOC: prim source of cashflow is sale of developed or improved properties rather than from recurring lease/rental ha

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(#) i) NOI / cap rate + cash / eqv. + Land for fut dev + Prepaid (#)

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Learning Module 3 | Investments in Real Estate Through Publicly Traded Securities

Net Asset Value Approach

(#) (#)

$$NAV\ per\ share = \frac{Value\ of\ operating\ estate + Value\ of\ other\ assets - Total\ debt\ and\ liabilities}{Number\ of\ shares\ outstanding}$$

but REIT sh prc ≠ NAV ps ; ∵ public equity mkt investo ascribe diff val than priv eq mkt investors.

If valuation of operating real estate is not provided:

$$Value\ of\ operating\ estate = \frac{NOI_1}{Cap\ rate}$$ (NOI₁ → future NOI)

Video: https://youtu.be/WncC3BZmfs8

$$NOI = Gross\ rental\ revenue - Estimated\ vacancy\ and\ collections\ loss - Operating\ expenses$$

Relative Value Approach

Funds from Operations:

$$FFO = Net\ income + Depreciation\ and\ amortization - Gains\ on\ sale\ of\ property + Loss\ on\ sale\ of\ property$$

Adjusted Funds from Operations: → subtracts recurring CapEx & the diff b/w reported vs cash re

$$AFFO = FFO - Non\ cash\ rent - Recurring\ capital\ expenditure\ and\ leasing\ costs$$

i.e. rent straight lining

Two-Stage Dividend Discount Model

$$Value\ of\ a\ REIT\ share = PV\ of\ dividends + PV\ of\ terminal\ value$$

Cash on-cash Ret^n = Cash after Interest / Equity ; may use debt to finance purchase

Hedonic index does not req repeat sales of same prop; only one sale. Regn controls for diff chars. of prop

Gross lease : owner pay OpEx ; Net Lease : tenant pays OpEx (eg. Triple Net : pay OpEx = CAM (common ar. Ma + Prop Taxes + bldg insur cost) ; Percent Lease : tenant pays addnl rent when sales hit a level.

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Both dedicated short sellers & short-biased mngrs will hold cash to moderate pfo β exposure.
↳ rely on single stock picking (v/s index) to achieve short exposr ; but short-biased may long the index.

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Learning Module 4 | Hedge Fund Strategies

Pairs trading on RV may be suited for specific stocks; index val may not be appropriate to examine ov/u val^n ?

Equity Market Neutral Pairs Trading

$$\text{Amount of Short Position in Overvalued Stock} = -\frac{\text{Beta of undervalued stock} \times \text{Amount Invested}}{\text{Beta of overvalued stock}}$$ OR $\beta_{UV} Am_{OV} = \beta_{OV} Am_{UV}$

Merger Arbitrage Strategy

Ret^n profile is exactly opposite b/w pers v/s fail of M&A deal
↳ ⇒ holding riskless bond + short binary Put option.

For a stock-for-stock deal:

$$Payoff\ if\ merger\ is\ successful = (N_A \times P_A) - (N_T \times P_T)$$

where:
N_A = Number of acquirer's shares to short sell
P_A = Share price of acquirer post announcement
N_T = Number of target's shares to buy
P_T = Share price of target post announcement

Conditional Factor Risk Model

$$R_{i,t} = \alpha_i + \beta_{i,1}(Factor\ 1)_t + \beta_{i,2}(Factor\ 2)_t + \cdots + \beta_{i,K}(Factor\ K)_t + \\ + D_t\beta_{i,1}(Factor\ 1)_t + D_t\beta_{i,2}(Factor\ 2)_t + \cdots + D_t\beta_{i,K}(Factor\ K)_t + (error)_{i,t}$$

where:
$R_{i,t}$ = Return of hedge fund i in period t
$\beta_{i,K}(Factor\ K)_t$ = Exposure to risk factor K for hedge fund i in period t during normal times
$D_t\beta_{i,K}(Factor\ K)_t$ = Incremental exposure to risk factor K for hedge fund i in period t during financial crisis periods
D_t = Dummy variable that equals 1 during financial crisis periods (0 otherwise)
α_i = Intercept for hedge fund i
$(error)_{i,t}$ = Random error with zero mean and standard deviation σ_i

Opportunistic :- take advtg of broader investm. opptst across various (A) classes ; not necessarily on Px ineff.
If mispricing is small, eg. govt secu, → may have to inc levrg to inc returns.
B/w Fund-of-fund & multi strat (or even overall), the F-o-F offers the highest level of liquidity.

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Rolling Window: backtest with rebal. aft each period & track pfmnc.
Creation/redemp^n is on-kind & does NOT trigger a tax event. // Tax lot mgmt within ETF to limit unrealzd CG.

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PORTFOLIO MANAGEMENT

VOLUME 5

Learning Module 1 | Exchange-Traded Funds: Mechanics and Applications

Arbitrage when M.Px. of ETF ≠ f.v. of ETF → viz per-share MP of ETF pfo holdings.

End-of-day ETF premium or discount (%)

$$\frac{ETF\ price - NAV\ per\ share}{NAV\ per\ share}$$

Intraday ETF premium or discount (%)

$$\frac{ETF\ price - Indicated\ NAV\ per\ share}{Indicated\ NAV\ per\ share}$$

✓ $Holding\ period\ cost\ (\%) = Round\ trip\ trade\ cost\ (\%) + Management\ fee\ (\%)$ → quoted annually → so may have to adjust for holding period to enable comps

✓ $Round\ trip\ trade\ cost\ \% = One\ way\ commission\ \% \times 2 + Bid\ ask\ spread\ \%$ → can be low due to :- 1) low creat^n redem fees, 2) pres. of hedging secu, 3) No ulty secu in other mkts 4) MKT Maker ↑ 5) Order Flow ↑

Learning Module 2: Using Multifactor Models

Arbitrage Pricing Theory (APT) : adding assets to pfo inc diversifica^n & asset-sp. risk eliminated

$$E(R_p) = R_F + \lambda_1\beta_{p,1} + \cdots + \lambda_K\beta_{p,K}$$

APT does not make assumpn wit mkt liquidity ; APT assumes investors can form well-diversf pfo that eliminate asset-sp

where:

$E(R_p)$ = the expected return to portfolio p

R_F = the risk-free rate

$\beta_{p,j}$ = the **sensitivity** of the **portfolio** to **factor** j

λ_j = the **expected reward** for bearing the risk of factor j

K = the **number of factors**

Carhart Four-Factor Model Not in excess of risk-free rate ; but APT states prem in excess of

$$E(R_p) = R_F + \beta_{p,1}RMRF + \beta_{p,2}SMB + \beta_{p,3}HML + \beta_{p,4}WML$$

where:

$RMRF$ = Return on a value-weighted equity index *minus* **one-month T-bill rate**

SMB = small minus big; **average return** on three **small-cap portfolios *minus*** the **average return** on three **large-cap portfolios** If β for SMB < 0 ⇒ large-cap firm

HML = high minus low; **average return** on two **high book-to-market portfolios *minus*** **average return** on two **low book-to-market portfolios**

WML = winners minus losers, a **momentum factor**; return on a portfolio of past year's winners minus return on a portfolio of past year's losers.

Factor Analy Models look to explain histor Ret^n Co-Variances. Factors are pfo of secu that explain hist ret

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factors obtained from ← ret" of ... securities in a group.

Macroeconomic Factor Model – easier to interpret vis statistic models but statistic fac Models make fewer assumptions than macro & funda Models. Stat Fac Model do not define facs up front. → PCA: explain histo ret" Var

$$R_i = a_i + b_{i1}F_1 + b_{i2}F_2 + \cdots + b_{iK}F_K + \varepsilon_i$$

In PCA, factors are pfs of secu that explain histo ret" Var.

where:

F_k = the **surprise in the factor** k; (Imp) develop the factor (surprise) series first & then estim fac. sensi via Regn

b_{ik} = the **sensitivity** of the return on asset i to a surprise in factor k, $k = 1, 2, \ldots,$

a_i = Expected return on the portfolio

Fundamental Factor Model – explain c/s diffn in stock prices

$$b_{ik} = \frac{\text{Value of attribute } k \text{ for asset } i - \text{Average value of attribute } k}{\sigma(\text{Values of attribute } k)} = \frac{A_i - \bar{A}}{\sigma}$$

∴ Added β for avg Val = 0

→ specify the fac sensi (attributes) first & then estim factor ret" via Regn.

Return Attribution

$$\text{Active return} = R_P - R_B = \sum_{k=1}^{K}\left[\begin{pmatrix}\text{Portfolio}\\ \text{sensitivity}\end{pmatrix}_k - \begin{pmatrix}\text{Benchmark}\\ \text{sensitivity}\end{pmatrix}_k\right] \times \begin{pmatrix}\text{Factor}\\ \text{return}\end{pmatrix}_k + \text{Security selection}$$

active exposure – shld align with invm mandate.

factor ret"

$$\text{Tracking error}, \quad TE = s(R_P - R_B)$$

$$\text{Information ratio}, \quad IR = \frac{\bar{R}_P - \bar{R}_B}{s(R_P - R_B)} = \frac{\text{Active return}}{\text{Tracking error}}; \quad \text{Sharpe Ratio} = \frac{R_P - R_F}{\sigma_P}$$

& $IR = IC \cdot TC \cdot \sqrt{BR}$, where IC = Info coeff ; TC = transfr coeff ; BR = breadth. // Funda Law of Activ Mgt @ S2

Active risk squared = Active factor risk + Active specific risk

→ way to measure consistency of active ret". ; SR measures pfo ret" in excess of riskless rate.

→ Sub-divide to security selection + allocat" @S2

2 Categories: A) Co. Funda → int pfmnce (emg mrkts, Euro (emg) Eq; B) Co.-share related – Val" Mult & trading shares, eg. P/E ; C) Macro factors; overall eco. pfmnce

Info Coeff (IC) = correl" b/w forecasted active ret" vis realized activ ret",

Transfr Coeff (TC) = measures the degree to which a fund mngr's forecast is translated into activ wts.

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convert annual s/dev to daily : s/dev = sdev/$\sqrt{T}$

pfo Transition Mgmt : hiring/firing pfo mngr OR making chngs to alloca^n with existing mngrs.

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Learning Module 3 | Measuring and Managing Market Risk

$t_{cr} = 1.645$ for 5% left tail

Parametric VaR (Using Normal Distribution)

$$\text{Value at Risk, } VaR = -[E(R_p) - z \times \sigma_p] \times \frac{\text{Portfolio}}{\text{Value}}$$

→ may convert freq.

where: Reliability of VaR can be verified thr back-testing (++ve)

$E(R_p)$ = Portfolio expected return

σ_p = Portfolio standard deviation ; $z = 1.65$ for 5% Var

VaR u/estim risk during extrm mkt stress ∵ it's sensi to rising correla^n eroding the diversfn benefit of normal

Two-asset portfolio:

$$E(R_p) = w_1 E(R_1) + w_2 E(R_2)$$

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1 w_2 \rho_{1,2}\sigma_1\sigma_2$$

Scaling from **daily** returns to **annual** returns (Assuming 1 year = 250 trading days):

$$R_{daily} \times 250 \text{ trading days}$$

Scaling from **daily** standard deviation to **annual** standard deviation:

$$\sigma_{daily} \times \sqrt{250}$$

Relative VaR = degree to which curr pfo pfmnc deviates from b/mk ; tracking risk w.r.t b/mk

$Incremental\ VaR\ (IVaR) = VaR\ after\ change - VaR\ before\ change$; CVar = Exp tail Loss /exp. shortfall → avg loss that wld be incurred if VaR cut off is exceeded. → conditional on exceeding VaR

Percentage change in bond price:

$$\frac{\Delta B}{B} \approx -Duration\frac{\Delta y}{1+y} + \frac{1}{2}Convexity\left(\frac{\Delta y}{1+y}\right)^2$$

New call price: $c + \Delta c \approx c + \text{Delta}_c(\Delta S) + \frac{1}{2}\text{Gamma}_c(\Delta S)^2 + Vega_c(\Delta\sigma)$

New put price: $p + \Delta p \approx p + \text{Delta}_p(\Delta S) + \frac{1}{2}\text{Gamma}_p(\Delta S)^2 + Vega_p(\Delta\sigma)$

Learning Module 4 | Backtesting and Simulation

No formula.

Histo simulation – VaR estim by repricing current pfo with actual histo chngs during lookback period.

MC simula^n :– mag. of losses & how freq.

Rev. stress testing – ID pfo. most sf exposure → push to extreme ⊆ hypoth scenario

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Real GDP grow rate ↑ => Indiv saving today ↓ ∵ higher exp. future incom due to more GFS in the future.
M.u. of fut consmpn ↓ ← greater fut consmpn vs today

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VOLUME 6 ~~ECONOMICS~~

Learning Module 1 | Economics and Investment Markets

One-period real-risk free rate:

$$l_{t,1} = \frac{1}{E_t[\tilde{m}_{t,1}]} - 1 \quad ; \quad E(m_t) = \frac{m_t}{m_0} = \frac{\text{MU consum in fut}}{\text{MU consum today}}$$

where:

$E_t[\tilde{m}_{t,1}]$ = Inter-temporal rate of substitution

$$\text{Price of risky asset} = \frac{E[\tilde{P}_{t+1,s-1}]}{1 + l_{t,1}} + cov_t[\tilde{P}_{t+1,s-1}, \tilde{m}_{t,1}]$$

(future Pxe)

-ve Cov => risky asset's payout ↓ when investor's (M) is hi → Pxe ↓ ∵ cov < 0 ∴ investors demand +ve risk prem to compensate.

as Cov( ) ↑ => risk prem ↑

where:

$\frac{E[\tilde{P}_{t+1,s-1}]}{1+l_{t,1}}$ = risk neutral present value

$cov_t[\tilde{P}_{t+1,s-1}, \tilde{m}_{t,1}]$ = covariance between investor's inter-temporal rate of substitution and the random future price the investment at t + 1, based on the information available to investor today.

s = time to maturity of investment

Asset with hi ret^n during bad times ; when MU of consmpn today is hi & ∴ provides hedge. ∴ Cov( ) is +ve

Default-free nominal coupon-paying bond

$$P_t^i = \sum_{s=1}^{N} \frac{CF_{t+s}^i}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s})^s}$$

where:

$l_{t,s}$ = Real-risk free rate

$\theta_{t,s}$ = Expected inflation rate

$\pi_{t,s}$ = Uncertainty in future inflation rate

$\theta_{t,s} + \pi_{t,s}$ = Breakeven rate of inflaton

Short-dated nominal zero-coupon government bonds (e.g., T-bills)

$$P_t^i = \frac{CF_{t+s}^i}{(1 + l_{t,s} + \theta_{t,s})^s}$$

S.T. bonds r valued more for paying in bad times (vs LT bonds) -> ∴ more reliable hedges. (∵ less +vely corr w bad times)

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If country below steady state of growth, it'll try to catch up → also characterised by higher real def-free int rate
Real GDP growth ∝ Real default-free int rate.

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Taylor Rule

$$pr_t = I_t + \pi_t + 0.5(\pi_t - \pi_t^*) + 0.5(Y_t - Y_t^*)$$

If actual GDP growth > potential GDP grw → inflat^n ↑ ⇒ Nom int rate↑ → bond Px↓

where:

pr_t = policy rate at time t

I_t = level of **real** short-term interest rates that balance long-term savings and borrowing in the economy

π_t = rate of inflation

π_t^* = **target** rate of inflation

Y_t = logarithmic level of **actual GDP**

Y_t^* = logarithmic level of **potential real GDP**

$Y_t - Y_t^*$ = output gap

Corporate bond

$$P_t^i = \sum_{s=1}^{N} \frac{E_t[\widetilde{CF}_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s})^s}$$

where:

$\gamma_{t,s}$ = Credit premium

Equity

$$P_t^i = \sum_{s=1}^{\infty} \frac{E_t[\widetilde{CF}_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s} + \kappa_{t,s})^s}$$

$$P_t^i = \sum_{s=1}^{\infty} \frac{E_t[\widetilde{CF}_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \lambda_{t,s})^s}$$

where:

$\kappa_{t,s}$ = Equity premium relative to risky bonds

$\lambda_{t,s} = \gamma_{t,s} + \kappa_{t,s}$ = Equity risk premium

Commercial Real Estate

$$P_t^i = \sum_{s=1}^{N} \frac{E_t[\widetilde{CF}_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s} + \kappa_{t,s} + \phi_{t,s})^s}$$

where:

$\phi_{t,s}$ = liquidity risk premium

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Learning Module 2 | Analysis of Active Portfolio Management

Active return, $R_A = R_P - R_B$

$Alpha, \alpha_p = R_p - \beta_p R_B$

$$Value\ added, \quad R_A = \sum_{j=1}^{M} \Delta w_j R_{B,j} + \sum_{j=1}^{M} w_{P,j} R_{A,j}$$

allocation | security selection

$SR_P^2 = SR_B^2 + IR^2$

$\sigma^2(R_P) = \sigma^2(R_B) + \sigma^2(R_A)$

For **optimal** Sharpe ratio,

$$\sigma(R_A) = \frac{IR}{SR_B}\sigma(R_B)$$

$$Transfer\ Coefficient, \quad TC = Corr\left(\frac{\mu_i}{\sigma_i}, \Delta w_i \sigma_i\right)$$

$$Information\ Coefficient, \quad IC = Corr\left(\frac{R_{Ai}}{\sigma_i}, \frac{\mu_i}{\sigma_i}\right)$$

$IC \approx 2(Probability\ of\ right\ call) - 1$

Forecasted active return, $\mu_i = IC \times \sigma_i \times S_i$

where: S_i is set of standardized forecasts of expected returns across securities

Mean-variance **optimal weights**

$$\Delta w_i^* = \frac{\mu_i \sigma_A}{\sigma_i^2 IC\sqrt{BR}}$$

Full Fundamental Law

& $IR = IC\sqrt{BR}$; BR = # Indep decisn

Max $E(R_A) = TC \times IC\sqrt{BR}\sigma_A$

use to Maxi val added in the choice of mgrs

$$IR = TC \times IC\sqrt{BR}$$

$$\sigma(R_A) = TC \times \frac{IR^*}{SR_B}\sigma(R_B)$$

SR maxi combi = index fund + active pfo

→ Corr btw fk ret^n & activ rets.

$$SR_P^2 = SR_B^2 + (TC)^2(IR^*)^2$$

Fundo Law: +ve : 1) forwards to compare mgrs on skill ; 2) operational measmt of elements of activ mgmt strat.

Limitn : 1) Does not consider Cov in calc. breadth. However, model can measure ex ante/post value added.

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Performance Measurement

$R_A = E(R_A|IC_R) + Noise$

TC^2 = Proportion of variation in realized performance attributed to realized information coefficient

where:

IC_R = realized information coefficient

Ex-ante measurement of skill

$$E(R_A) = \frac{IC}{\sigma_{IC}}\sigma_A$$

Independence of Investment Decision

$$BR = \frac{N}{1 + (N-1)\rho}$$

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